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Master Thesis

**Self-Supervised Learning of Latent EEG
Representations for Mental Health
Related Analysis**

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**Master of Science
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Abstract (en)

Resting-state electroencephalography (EEG) provides a non-invasive measure of brain activity relevant to neurological and psychiatric conditions. However, substantial variability across participants and independently collected datasets limits the generalisability of learned representations. Self-supervised learning offers a way to learn reusable representations directly from unlabelled EEG, yet its utility for resting-state clinical classification and the factors that influence generalisation remain unclear.

This thesis adapts BrainLM, a masked autoencoder originally developed for functional magnetic resonance imaging, to multichannel EEG. Two novel variants investigate complementary representation learning choices: BrainLM-EEG-RoPE incorporates electrode geometry through rotary positional encoding, while BrainLM-EEG-Microstate introduces an auxiliary EEG microstate prediction objective. The models were pretrained on resting-state EEG from six datasets comprising 925 participants and evaluated across four clinical classification tasks against classical, supervised, and other existing pretrained EEG models. Generalisation was assessed both to unseen participants within a dataset and to independently collected data.

The proposed models learned reusable representations that were competitive across clinical tasks and supported strong participant level classification in several settings. Their downstream utility varied with the clinical task, representation design, and evaluation protocol, highlighting the importance of these design choices in self-supervised EEG modelling. Cross-dataset evaluation further revealed a clear distinction between participant level generalisation and robustness to dataset shift, with external transfer remaining difficult across modelling approaches. Overall, the findings support masked self-supervised learning as a viable framework for resting-state EEG representation learning and highlight representation design and external validation as key considerations for developing more generalisable clinical EEG models.

Abstract (de)

Die Ruhezustands-Elektroenzephalographie (EEG) ermöglicht die nicht-invasive Erfassung von Hirnaktivität, die für die Untersuchung neurologischer und psychiatrischer Erkrankungen von Bedeutung ist. Eine ausgeprägte Variabilität zwischen Teilnehmenden sowie zwischen unabhängig erhobenen Datensätzen erschwert jedoch die Generalisierbarkeit gelernter Repräsentationen. Self-Supervised Learning bietet die Möglichkeit, wiederverwendbare Repräsentationen unmittelbar aus ungelabelten EEG-Daten zu erlernen. In welchem Umfang solche Repräsentationen für die klinische Klassifikation auf Basis von Ruhezustands-EEG geeignet sind und welche Faktoren ihre Generalisierbarkeit beein-

flussen, ist bislang nicht abschließend geklärt.

In dieser Arbeit wird BrainLM, ein ursprünglich für die funktionelle Magnetresonanztomographie entwickelter Masked Autoencoder, auf mehrkanaliges EEG übertragen. Zwei neu entwickelte Varianten untersuchen komplementäre Ansätze des Repräsentationslernens: BrainLM-EEG-RoPE berücksichtigt die räumliche Elektrodengeometrie mittels Rotary Positional Encoding, während BrainLM-EEG-Microstate eine zusätzliche Vorhersage von EEG-Mikrozuständen in die Lernaufgabe integriert. Die Modelle wurden auf Ruhezustands-EEG aus sechs Datensätzen mit insgesamt 925 Teilnehmenden vortrainiert und in vier klinischen Klassifikationsaufgaben mit klassischen Verfahren, überwacht trainierten Modellen sowie weiteren vortrainierten EEG-Modellen verglichen. Die Generalisierbarkeit wurde sowohl für bislang ungesehene Teilnehmende innerhalb eines Datensatzes als auch anhand unabhängig erhobener Daten untersucht.

Die vorgeschlagenen Modelle erlernten wiederverwendbare Repräsentationen, die über verschiedene klinische Aufgaben hinweg konkurrenzfähig waren und in mehreren Evaluationsszenarien eine hohe Klassifikationsleistung auf Teilnehmendenebene ermöglichten. Ihre Nutzbarkeit für nachgelagerte Aufgaben variierte in Abhängigkeit von der klinischen Aufgabe, dem Repräsentationsdesign und dem Evaluationsprotokoll und verdeutlicht damit die Bedeutung dieser Designentscheidungen für Self-Supervised Learning mit EEG. Die datensatzübergreifende Evaluation zeigte zudem einen deutlichen Unterschied zwischen der Generalisierung auf ungesehene Teilnehmende und der Robustheit gegenüber Datensatzverschiebungen, wobei der externe Transfer über verschiedene Modellierungsansätze hinweg anspruchsvoll blieb. Insgesamt stützen die Ergebnisse Masked Self-Supervised Learning als einen tragfähigen Ansatz zum Erlernen wiederverwendbarer Repräsentationen aus Ruhezustands-EEG und unterstreichen die Bedeutung des Repräsentationsdesigns und externer Validierung für die Entwicklung besser generalisierbarer klinischer EEG-Modelle.

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1 Introduction

Resting-state electroencephalography (EEG) provides a non-invasive measure of ongoing brain activity and is widely used to investigate neural patterns associated with neurological and psychiatric conditions. However, EEG exhibits substantial inter-participant variability, while recordings acquired across datasets may additionally differ in cohort composition, electrode configuration, acquisition hardware, sampling rate, preprocessing, and recording conditions. These sources of variability can limit the extent to which representations learned from one dataset remain informative for unseen participants or independently collected data (Kim et al., 2023; Engemann et al., 2021; Wagh, Wei et al., 2022). Participant level and cross-dataset generalisation are therefore important challenges in clinical EEG analysis.

Conventional EEG classification approaches typically rely on handcrafted features combined with classical machine learning classifiers or on supervised deep learning models trained directly on the downstream task (Craik et al., 2019; Gemein et al., 2020). More recently, self-supervised learning (SSL) has been used to pretrain EEG encoders using objectives derived from the signal itself, after which the learned representations are transferred to downstream classification tasks (Banville et al., 2020; Kostas et al., 2021).

Recent studies have shown that self-supervised pretraining can produce EEG representations that support downstream classification across a range of tasks (Jiang et al., 2024; J. Wang et al., 2025; El Ouahidi et al., 2026). However, the effectiveness of these approaches for resting-state clinical EEG remains less established. In particular, it remains unclear how self-supervised representations compare with classical and supervised approaches across different clinical tasks, and under what conditions they provide useful transferable representations. Moreover, successful generalisation to unseen participants within a dataset does not necessarily imply robust transfer to independently collected datasets, where additional differences in cohorts and acquisition conditions are introduced.

The utility of self-supervised EEG representations may also depend on choices made during pretraining and representation extraction, including masking configuration, spatial encoding, auxiliary objectives, and encoder depth. For multichannel EEG, learned channel embeddings encode channel identity but do not explicitly capture the physical relationships between electrodes, motivating the use of geometry-aware spatial representations. EEG microstates provide a complementary description of recurring whole

scalp topographies that have been associated with group level differences in neurological and psychiatric conditions (Khanna, Pascual-Leone and Farzan, 2014; Koenig et al., 2023). Whether explicit electrode geometry or microstate-informed auxiliary supervision improves downstream performance within a common self-supervised framework, and how these effects interact with masking and representation depth, therefore remains an open empirical question. These design choices are examined in greater detail in Section 3.7.

In this thesis, BrainLM (Caro et al., 2024), a masked autoencoder framework originally developed for functional magnetic resonance imaging (fMRI), was adapted to resting-state EEG. The resulting BrainLM-EEG model represents multichannel EEG as channel-time patches and is pretrained by reconstructing masked signal content. Two novel variants were developed to investigate specific choices in self-supervised EEG representation learning. BrainLM-EEG-RoPE replaces learned spatial channel embeddings with geometry-aware rotary positional encoding derived from electrode locations, while BrainLM-EEG-Microstate augments the reconstruction objective with an auxiliary EEG microstate prediction task. The remaining architecture and pretraining framework are kept consistent across the variants, allowing the effect of each modification to be examined separately.

The empirical evaluation comprises four clinical classification tasks: Alzheimer’s disease (AD) versus healthy controls, frontotemporal dementia (FTD) versus healthy controls, first-episode psychosis (FEP) versus healthy controls, and major depressive disorder (MDD) versus healthy controls. BrainLM-EEG and its variants are compared with classical machine learning approaches, supervised deep learning models, and existing pretrained EEG models using both linear probing and finetuning. Participant level generalisation is evaluated using five-fold cross-validation and leave-one-subject-out (LOSO) evaluation, while cross-dataset generalisation is assessed by transferring models trained on the FEP dataset to an independently collected schizophrenia (SCZ) dataset. Additional experiments examine masking configuration, encoder depth, the contribution of the CLS token to masked reconstruction, and the downstream information retained in reconstructed EEG.

1.1 Research Questions

The primary goal of this thesis is to determine whether the proposed self-supervised BrainLM-EEG models learn useful and generalisable representations of resting-state EEG for clinical classification. A further goal is to determine how selected pretraining and architectural choices affect downstream performance and to investigate what information is used and retained by the learned representations. These goals are addressed through the following research questions:

1. **RQ1:** How do the proposed BrainLM-EEG models compare with classical machine learning, supervised deep learning, and existing pretrained EEG models on downstream clinical classification tasks, and how does performance differ between linear probing and finetuning?
2. **RQ2:** How well do the proposed BrainLM-EEG models generalise to unseen participants within a dataset and to an independently collected dataset, relative to supervised baselines?
3. **RQ3:** How do design choices including masking configuration, spatial positional encoding, auxiliary pretraining objectives, and encoder depth affect downstream classification performance?
4. **RQ4:** What do CLS token perturbation and classification of reconstructed EEG reveal about the information encoded and retained by BrainLM-EEG?

Together, these questions assess the downstream utility of the proposed representations, their generalisation across participants and datasets, the influence of selected representation learning choices, and the information retained during self-supervised pretraining.

1.2 Contributions

The main contributions of this thesis are:

- **BrainLM-EEG:** This thesis adapts BrainLM, a masked autoencoder originally developed for fMRI, to multichannel resting-state EEG. The resulting model is pretrained on 925 participants from six datasets.
- **Two extensions:** Two novel variants are developed: BrainLM-EEG-RoPE incorporates electrode geometry through rotary positional encoding, while BrainLM-EEG-Microstate incorporates EEG microstates through an auxiliary prediction objective. Both extensions are evaluated under controlled conditions to isolate their respective contributions.
- **Controlled comparative evaluation:** BrainLM-EEG and its variants are evaluated alongside classical machine learning methods, supervised deep learning models, and existing pretrained EEG models across four clinical classification tasks. Evaluation uses participant level data splits and a common protocol designed to prevent participant leakage during pretraining and downstream evaluation.

- **Systematic evaluation of representation and generalisation:** The thesis investigates the effects of masking configuration and encoder depth on downstream performance and evaluates generalisation under both participant level cross-validation and external cross-dataset transfer. Both linear probing and fine-tuning are evaluated to assess how the learned representations transfer to clinical tasks.

1.3 Thesis Outline

The subsequent chapters are organised as follows. Chapter 2 provides background on resting-state EEG, the clinical conditions considered in this thesis, and supervised and self-supervised representation learning. Chapter 3 reviews related work on EEG representation learning and clinical classification. Chapter 4 describes the datasets, proposed models, comparison baselines, pretraining procedure, downstream evaluation protocols, and model analysis experiments. Chapter 5 presents the experimental results. Chapter 6 discusses the findings, limitations, and implications and answers the research questions stated above. Chapter 7 concludes the thesis and outlines directions for future work.

2 Background

This chapter introduces the technical and clinical concepts required to understand the methods and experiments in this thesis, including resting-state EEG, the clinical conditions considered, self-supervised representation learning, EEG microstates, and positional encoding.

2.1 Electroencephalography

2.1.1 EEG Basics

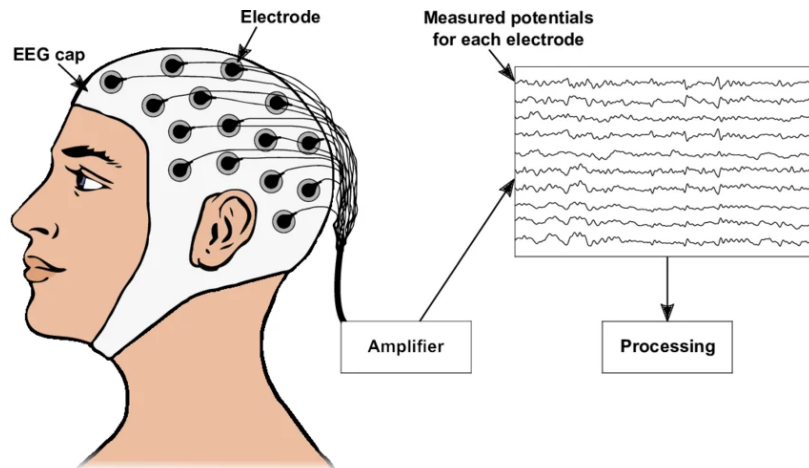


Figure 2.1: EEG recording chain from scalp electrodes to digitised time series. Reproduced from Nagel (2019).

Electroencephalography (EEG) is a non-invasive method for recording electrical brain activity from electrodes placed on the scalp. It provides high temporal resolution, allowing changes in brain activity to be measured at the millisecond scale. The recorded voltages mainly reflect the combined postsynaptic activity of large populations of cortical neurons. As these electrical fields travel through brain tissue, cerebrospinal fluid, skull, and scalp before reaching the electrodes, each EEG channel reflects a mixture of activity from multiple neural sources rather than a single isolated source (Jackson and Bolger, 2014). In

practice the scalp potentials are picked up by electrodes held against the head, amplified, and digitised, giving one time series per electrode for subsequent analysis, as shown in Figure 2.1.

Electrodes are commonly positioned using the international 10-20 system (Homan et al., 1987), which defines reproducible locations over the scalp. Electrode positions are defined relative to standard anatomical landmarks on the head, ensuring consistent placement across participants. Figure 2.2 shows these landmarks and the denser 10-10 extension, within which the 19 electrodes used throughout this thesis (Section 4.3.1) are located. The set of recorded electrodes and the reference scheme together define the montage. The reference is important because EEG measures voltage differences rather than absolute voltage. Changing the reference therefore changes the recorded channel values and can affect relationships between channels (Bigdely-Shamlo et al., 2015).

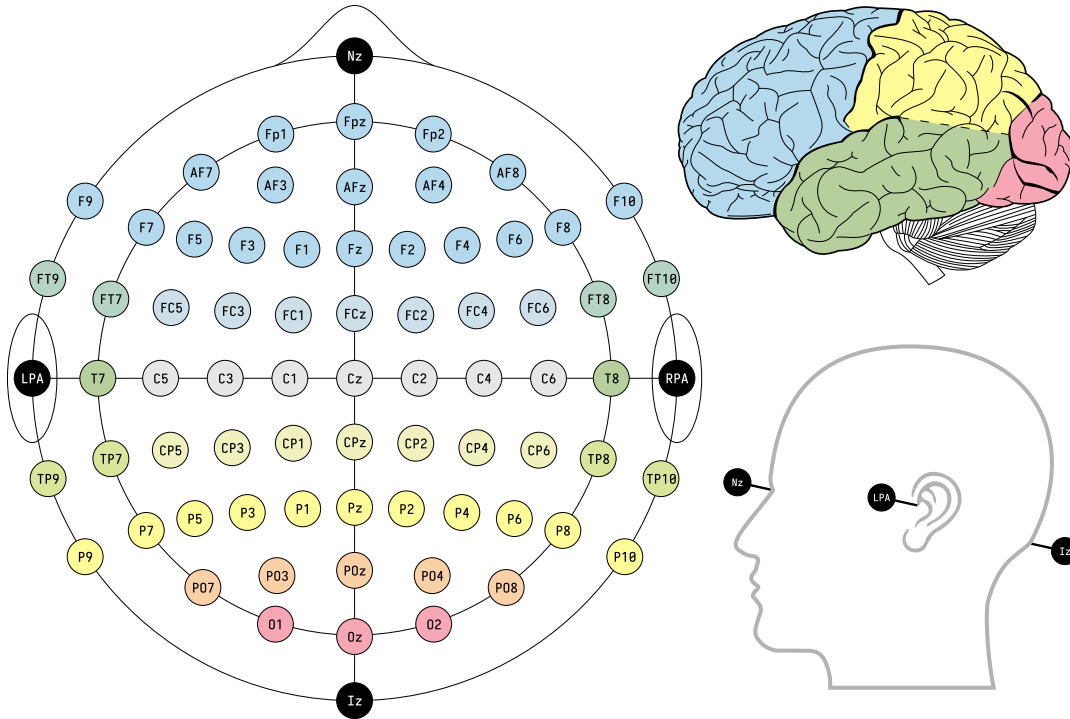


Figure 2.2: Electrode positions of the international 10-10 system, an extension of the 10-20 system. Labels follow the modified combinatorial nomenclature. Figure by Laurens R. Krol, released into the public domain under CC0 1.0 (Krol, 2020).

EEG is therefore a multichannel signal with an inherent spatial structure. Channels are usually stored in a fixed numerical order, but this order does not describe the physical distances or directions between electrodes on the scalp. Channel identity and electrode position are related but distinct forms of information, a distinction that becomes relevant when EEG is represented as a sequence for computational models (J. Wang et al., 2025; El Ouahidi et al., 2026).

2.1.2 Resting-State EEG and Frequency Bands

Resting-state EEG is recorded while no explicit task is being performed, usually with the participant relaxed and the eyes open or closed. The two conditions are not equivalent because eye closure commonly increases posterior alpha activity and changes the distribution of resting-state rhythms (Barry et al., 2007). Resting-state EEG is useful in clinical research because it requires little task understanding or behavioural response.

EEG is often described through approximate frequency bands: delta at about 1-4 Hz, theta at 4-8 Hz, alpha at 8-13 Hz, beta at 13-30 Hz, and gamma above about 30 Hz. These ranges are conventions rather than fixed physiological divisions, but they provide an interpretable summary of how signal power is distributed across frequencies (Teplan, 2002).

Frequency information is clinically useful because some neurological and psychiatric conditions are associated with changes in the balance between slower and faster rhythms. Alzheimer's disease (AD), for example, is commonly associated with relatively more slow activity and reduced alpha activity than healthy ageing (Jeong, 2004; Jelic et al., 1996). Frequency-band summaries are interpretable, but they average activity over time and do not fully describe transient changes or relationships between electrodes.

2.1.3 EEG Artefacts and Preprocessing

Scalp EEG contains neural activity together with unwanted signals from the eyes, muscles, heart, electrodes, movement, and the electrical environment. Eye blinks can produce large slow deflections, muscle activity can add broadband high-frequency energy, and power-line interference can introduce narrow-band noise. These artefacts can be larger than the neural activity of interest (Delorme and Makeig, 2004; Bigdely-Shamlo et al., 2015).

Preprocessing aims to reduce these effects and make recordings more comparable. Typical steps include filtering, re-referencing, detecting bad channels, and removing strongly contaminated components. Independent component analysis (ICA; Hyvärinen and Oja, 2000), for example, separates the observed signals into components with different statistical patterns, allowing components associated with eye, cardiac, or muscle activity to be identified and removed (Li et al., 2022).

Preprocessing is not a neutral cleaning step. Filtering changes the frequency content of the signal, re-referencing changes spatial relationships between channels, and artefact removal can also remove genuine neural activity. Differences in preprocessing and recording hardware can therefore influence the statistical properties of EEG data and complicate

comparisons across datasets (Bigdely-Shamlo et al., 2015; Roy et al., 2019).

2.2 Clinical Conditions

The clinical conditions considered in this thesis are Alzheimer’s disease (AD), frontotemporal dementia (FTD), first-episode psychosis (FEP), schizophrenia (SCZ), and major depressive disorder (MDD). They differ substantially in their symptoms, disease course, and underlying pathology. Diagnosis is primarily based on clinical assessment, supported where appropriate by cognitive testing, imaging, laboratory investigations, or other biomarkers. Resting-state EEG is not a stand-alone diagnostic test for these conditions yet, but it provides a non-invasive measure of brain function that can reveal group level differences in neural activity (Roy et al., 2019; van der Vinne et al., 2017).

2.2.1 Alzheimer’s Disease and Frontotemporal Dementia

AD and FTD are neurodegenerative disorders that progressively affect cognition and everyday functioning. AD most commonly begins with impairment of memory and gradually affects other cognitive abilities. FTD primarily affects frontal and temporal brain systems and often presents with early changes in behaviour, personality, executive function, or language. Clinical diagnosis is based on the pattern and progression of symptoms together with cognitive and neurological assessment, often supported by additional investigations (Miltiadous, Tzamourta et al., 2023).

Resting-state EEG in AD commonly shows spectral slowing, including increased relative delta and theta activity, reduced alpha activity, and a lower dominant frequency (Jeong, 2004; Jelic et al., 1996). EEG findings in FTD are less uniform, although changes in functional connectivity and network organisation have been reported (de Haan et al., 2009; Yu et al., 2016).

2.2.2 First-Episode Psychosis and Schizophrenia

Psychosis can involve hallucinations, delusions, and disturbances in thought or behaviour. FEP refers to the first clinical presentation of psychotic symptoms. Schizophrenia is a longer-term psychiatric disorder in which psychotic symptoms can occur together with negative symptoms, such as reduced motivation or emotional expression, and cognitive difficulties involving functions such as attention and memory. Diagnosis is based on

the nature, duration, and functional impact of these symptoms rather than on an EEG measurement (Uhlhaas and Singer, 2010).

EEG studies of schizophrenia report alterations in neural oscillations and in the synchronisation of activity across brain regions, although the direction and magnitude of these findings vary across cohorts and recording conditions (Uhlhaas and Singer, 2010). Differences have also been reported in the temporal variability of resting-state spectral activity (Racz et al., 2025).

2.2.3 Major Depressive Disorder

MDD is a psychiatric disorder characterised by persistent depressed mood, reduced interest or pleasure, or both, together with additional cognitive and physical symptoms that interfere with daily functioning. These can include changes in sleep, energy, concentration, appetite, and psychomotor activity. Diagnosis is based on the pattern, duration, and severity of symptoms and their effect on functioning (van der Vinne et al., 2017).

Resting-state EEG research in MDD has examined spectral power, functional connectivity, and frontal alpha asymmetry. Frontal alpha asymmetry compares alpha activity over left and right frontal regions, but evidence for a reliable diagnostic difference between MDD and healthy controls is weak and inconsistent across studies and analysis choices (van der Vinne et al., 2017; Kołodziej et al., 2021).

2.3 Supervised Learning for EEG

Supervised learning uses examples for which the target label is known. In clinical EEG classification, a model receives EEG data together with a diagnostic group label and learns a mapping from the signal to that label. Classical approaches first convert EEG into hand-designed features, whereas deep models learn features directly from the signal (Craik et al., 2019; Roy et al., 2019).

2.3.1 Handcrafted EEG Features and Classical Models

A traditional EEG pipeline summarises each recording window with a feature vector. The selected features describe properties expected to be informative, such as frequency content, waveform shape, temporal variability, or relationships between electrodes. This

reduces a high-dimensional time series to quantities that are easier to interpret and can work well when the number of participants is limited.

Spectral features. Band power measures how much signal power lies within a selected frequency range. It is especially useful when a clinical effect changes the balance of slow and fast activity, as in AD (Jeong, 2004). Its main limitation is that averaging power over a window reduces temporal detail.

Time-frequency and time-domain features. Time-domain features describe waveform properties such as variance or Hjorth activity, mobility, and complexity. Time-frequency methods describe how frequency content changes over time. Relationships between electrodes can additionally be represented through covariance, correlation, or coherence (Hjorth, 1970; de Haan et al., 2009). These families provide complementary views of the signal.

Classical classifiers. Feature vectors can be used with classifiers such as linear discriminant analysis (LDA), support vector machines (SVMs), logistic regression, or gradient-boosted trees such as XGBoost (T. Chen and Guestrin, 2016). These models remain important baselines because they are relatively data efficient. Their main limitation is that the feature set determines what information is available to the classifier.

2.3.2 Deep Learning for EEG

Deep neural networks learn feature extraction and classification jointly. Convolutional neural networks (CNNs) can learn local temporal patterns and combine information across electrodes, while transformer-based models use self-attention to model relationships between input segments over longer ranges (Craik et al., 2019; Song et al., 2023). In EEG, the input can be divided into short channel-time segments so that the model learns from both temporal and spatial context.

Supervised deep models learn their internal representation from labelled examples. In clinical EEG, labels can be expensive to obtain and the number of independent participants is often much smaller than the number of signal segments. Self-supervised learning provides an alternative way to learn representations from unlabelled recordings before they are used for a downstream task.

2.4 Self-Supervised Representation Learning

Self-supervised learning (SSL) creates a training target from the data itself rather than requiring a clinical label. An encoder is first trained on unlabelled EEG to solve a pretraining task. The encoder converts the input into numerical features, forming a learned representation that can later be used for labelled downstream tasks (Banville et al., 2020; Kostas et al., 2021).

2.4.1 Self-Supervised Learning Objectives

A self-supervised objective determines what information the encoder is encouraged to preserve. Contrastive methods train related signal segments or transformed views to have similar representations while separating unrelated examples. Reconstruction-based methods hide or corrupt part of the input and train the model to predict the missing information from what remains (Banville et al., 2020; Baevski et al., 2022).

For EEG, the objective must be chosen carefully because nearby time samples and neighbouring channels are strongly related. A task that is too easy may be solved by local interpolation, while a task dominated by noise may encourage the model to reconstruct artefacts rather than useful brain activity. The pretraining objective therefore shapes what information becomes available in the learned representation.

2.4.2 Masked Autoencoders

A masked autoencoder (MAE; He et al., 2022) divides an input into patches, masks a subset of them, and reconstructs the missing content from the remaining visible patches. The encoder processes only the visible patches, while the decoder receives their latent representations together with mask tokens and positional information to reconstruct the masked patches. The reconstruction objective encourages the encoder to capture dependencies between visible and masked parts of the input that are useful for inferring missing information.

The masked reconstruction principle can also be applied to sequential and multivariate signals, including recordings of brain activity. The representations learned through this process are influenced by how the signal is partitioned into patches, which patches are masked, and how positional or spatial information is encoded. These choices are particularly relevant for multichannel EEG, where information varies across both time and electrode location. Chapter 3 reviews how existing EEG representation learning models address these design choices.

2.5 EEG Microstates

Frequency bands describe how signal power is distributed across frequencies, but they do not directly describe the shape of the electrical field over the scalp at a particular moment. EEG microstates provide a complementary description. A microstate is a short period during which the spatial pattern of scalp voltages remains approximately stable before changing to another pattern (Pascual-Marqui et al., 1995; Michel and Koenig, 2018).

Microstate analysis reduces continuous multichannel EEG to a sequence of recurring scalp maps. Representative topographies are identified and then matched back to the recording, allowing the sequence to be described by properties such as duration, occurrence, coverage, and transitions (Pascual-Marqui et al., 1995; Férat et al., 2022). Common labels such as A, B, C, and D are names for recurring topographic patterns rather than fixed brain regions or disease classes. Their exact form depends on preprocessing and template estimation (Férat et al., 2022; Khanna, Pascual-Leone and Farzan, 2014). Figure 2.3 shows the four maps estimated for this thesis, which are used as prediction targets by one of the model variants introduced in Section 4.3.2.

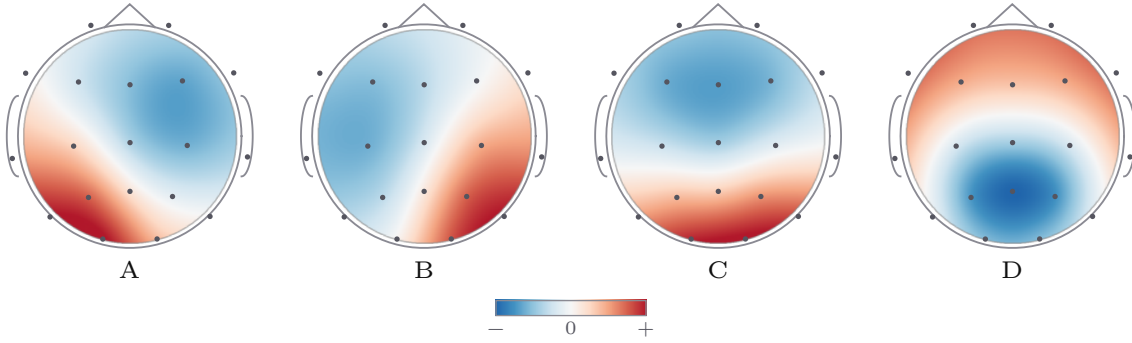


Figure 2.3: The four canonical microstate maps (A-D) used in this thesis, estimated from the LEMON training partition (Section 4.3.2). Each map is defined by one value at each of the 19 electrodes (dots); the continuous colour field between electrodes is obtained by spatial interpolation for visualisation. Colour indicates relative scalp potential from negative (blue) to positive (red).

2.6 Positional Encoding in Transformer Models

Self-attention (Vaswani et al., 2017) does not inherently encode the order or position of input tokens, so Transformer models require an additional mechanism for representing positional information (Dufter et al., 2022). For EEG, positional information can have both temporal and spatial components: temporal position describes when a patch occurs

within the recording, while spatial position describes the location of its corresponding electrode on the scalp.

A learned channel embedding can distinguish electrodes within a fixed montage, but it represents channel identity as a learned label rather than explicitly encoding the physical relationships between electrode locations. This distinction becomes relevant when electrode configurations differ across datasets, because learned channel embeddings are tied to the channel identities encountered during training and do not explicitly represent relative electrode geometry. Coordinate-based encodings instead represent electrodes using their spatial coordinates rather than their position in an arbitrary channel ordering. Rotary positional embedding (RoPE; Su et al., 2024) incorporates positional information into self-attention by applying position-dependent rotations to query and key representations, such that their interaction reflects relative positional relationships. RoPE has also been extended to multidimensional coordinates (Heo et al., 2025), allowing spatial relationships to be represented in more than one dimension.

3 Related Work

Building on the concepts introduced in Chapter 2, this chapter reviews prior work most closely related to the thesis, focusing on clinical EEG classification, self-supervised EEG representation learning, spatial modelling, microstates, and generalisation. The chapter concludes by identifying the research gaps addressed in this work.

3.1 Conventional EEG Analysis and Clinical Classification

Comparisons between feature-based and end to end approaches show that learned representations do not necessarily outperform well-designed handcrafted EEG features in clinical classification (Craik et al., 2019; Gemein et al., 2020). This makes conventional feature-based methods an important reference point when evaluating whether self-supervised representations provide additional value.

Gemein et al. (2020) compared a comprehensive feature-based framework with end to end neural networks for distinguishing pathological from non-pathological EEG. Despite their different approaches to feature extraction, the feature-based and neural models achieved similar classification performance. Analyses of their decision patterns further indicated that both approaches relied on similar low-frequency information, particularly delta and theta activity at temporal electrodes. These findings demonstrate the importance of including strong handcrafted baselines when evaluating whether learned representations provide information beyond established EEG descriptors.

Supervised neural architectures provide an additional comparison with self-supervised pretrained representations. EEGNet (Lawhern et al., 2018) uses a compact convolutional architecture with depthwise and separable convolutions designed for EEG feature extraction. Building on EEGNet, EEGNeX (X. Chen et al., 2024) introduces additional convolutional processing, an inverted-bottleneck structure, and dilated convolutions to enhance spatial feature extraction and capture temporal dependencies over a larger receptive field. EEG Conformer (Song et al., 2023) combines convolutional feature extraction with self-attention to model local features and longer-range dependencies. ATCNet (Altaheri et al., 2023) combines convolutional processing, multi-head self-attention, and temporal convolutional networks within a sliding window framework to model temporal structure

with a relatively small number of parameters. Although these architectures were evaluated mainly on brain-computer interface (BCI) and other task-based EEG paradigms, they provide useful supervised comparators for assessing whether self-supervised pretraining offers an advantage over representations learned directly from labelled data.

Clinical EEG studies also highlight the importance of evaluation design. DICE-Net (Miltiadous, Gionanidis et al., 2023) combined relative band power and spectral coherence features with a convolutional-transformer architecture for dementia classification and evaluated performance using leave-one-subject-out validation. In schizophrenia classification, Shoeibi et al. (2021) reported high cross-validation performance using CNN-LSTM models on a cohort of 14 patients and 14 healthy controls. However, results obtained from such small cohorts provide limited evidence of generalisation to independent participants or datasets. More broadly, these studies illustrate that high internal classification performance alone is insufficient to establish robustness in clinical EEG, evaluation must also account for participant independence and, where possible, transfer to independently collected data.

3.2 Self-Supervised Learning for EEG

Prior self-supervised EEG studies have investigated predictive, contrastive, and reconstruction based objectives (Weng et al., 2024). Early clinical EEG work demonstrated that the design of the pretraining task strongly affects the information learned. Banville et al. (2020) compared relative positioning, temporal shuffling, and contrastive predictive coding and evaluated the pretrained encoders with frozen linear classifiers. The learned features were particularly useful when few labels were available and captured information related to age, sleep stage, and pathology. However, the choice of negative examples also changed what the representation encoded: sampling negatives across recordings encouraged sensitivity to differences between recordings. This finding is important because self-supervised objectives can learn stable participant or recording characteristics as well as reusable physiological structure.

BENDR (Kostas et al., 2021) used a convolutional encoder and transformer to learn contextual representations by predicting latent future features among distractors. It was pretrained on a large clinical EEG corpus and transferred across several EEG tasks and recording setups. The results provided evidence that large-scale self-supervised pretraining can produce reusable EEG features, although most downstream tasks were not resting-state psychiatric or neurodegenerative classification.

Masked reconstruction provides a different form of self-supervision. MAEEG (Chien et al., 2022) masked latent EEG features and reconstructed the signal with a transformer-based decoder. In sleep staging experiments, more difficult masking configurations

and longer input contexts produced stronger downstream representations than easier reconstruction tasks. EEG2Rep (Foumani et al., 2024) instead predicted latent target representations and used structured masking intended to preserve coherent context. Its evaluation included linear probing, finetuning, and transfer across EEG tasks. These studies suggest that masking ratio, mask geometry, input duration, and prediction target all influence the representations learned from EEG rather than acting as purely technical hyperparameters.

3.3 Pretrained Self-Supervised EEG Models

Recent work has scaled self-supervised pretraining to larger and more heterogeneous EEG collections. These models differ in tokenisation, pretraining objectives, treatment of electrode information, and downstream adaptation (Kuruppu et al., 2025). Their common goal is to learn transferable representations rather than training a new encoder from scratch for each task.

BrainLM (Caro et al., 2024) provides a closely related example from functional magnetic resonance imaging (fMRI) and forms the architectural basis of this thesis. It represents regional brain activity as temporal patches and uses masked autoencoding to reconstruct missing signal content from visible patches. BrainLM was evaluated on clinical variable prediction and external-cohort generalisation. However, fMRI and EEG differ substantially in their spatial and temporal characteristics, BrainLM operates on regional fMRI time series, whereas EEG consists of rapidly sampled signals recorded at scalp electrodes. Section 4.3.1 describes how BrainLM is adapted to EEG in this thesis.

Several self-supervised EEG models address signal heterogeneity in different ways. BIOT (Yang et al., 2023) tokenises channels independently and combines spectral features with channel and temporal embeddings. LaBraM (Jiang et al., 2024) learns discrete neural codes for EEG patches and predicts the codes of masked patches. EEGPT (G. Wang et al., 2024) combines masked waveform reconstruction with spatiotemporal representation alignment and learned channel mappings. EEGFormer (Y. Chen et al., 2024) also learns discrete EEG representations and evaluates transfer from a large clinical corpus to an independent neonatal seizure dataset.

Other approaches focus more explicitly on spatial and temporal structure. CBraMod (J. Wang et al., 2025) uses criss-cross attention to model spatial and temporal dependencies separately, while REVE (El Ouahidi et al., 2026) incorporates three-dimensional electrode coordinates and temporal position together with spatiotemporal masking. These approaches illustrate different assumptions about how EEG should be represented during pretraining. However, existing evaluations remain heterogeneous, and current evidence does not establish a single representation or pretraining strategy that is consistently

superior across datasets and downstream tasks (Kuruppu et al., 2025). This motivates controlled comparisons in which representation choices are varied while the underlying architecture, pretraining data, and downstream evaluation protocol are kept as consistent as possible.

3.4 Spatial and Geometric Inductive Biases

Because EEG channels correspond to physical sensor locations, several approaches explicitly model relationships between electrodes. Graph-based methods represent electrodes as nodes and define connections using spatial distance, functional connectivity, or both. EEG-GCNN (Wagh and Varatharajah, 2020) combined the geometry of a 10-20 montage with spectral coherence for neurological disease classification. The study demonstrated the potential value of spatial structure, although recordings from clinical and healthy groups were acquired using different systems, making acquisition differences a possible confound.

Other methods attempt to handle montage variation without defining a fixed graph. Yi et al. (2023) mapped different electrode configurations to a common representation, combining multidimensional positional encoding with a channel hierarchy. BIOT (Yang et al., 2023) used learned channel embeddings, EEGPT (G. Wang et al., 2024) used channel mappings, and REVE (El Ouahidi et al., 2026) derived positional information directly from electrode coordinates. These methods encode different assumptions: channel embeddings represent sensor identity, topology mappings align electrodes to a common structure, and coordinate-based methods preserve continuous physical location.

Rotary positional embedding (RoPE; Su et al., 2024), introduced in Section 2.6, provides another mechanism for representing position in Transformer attention. RoPE was originally developed for sequential positions rather than multidimensional electrode coordinates, so its application to EEG sensor geometry requires adapting the positional representation. Existing coordinate-based EEG models nevertheless suggest that explicit sensor geometry can be useful across varying electrode configurations (El Ouahidi et al., 2026). The literature therefore motivates comparison between learned channel identity and geometry-aware spatial representations without assuming that physical proximity alone reflects neural connectivity.

3.5 EEG Microstates in Clinical Populations

Prior microstate research, building on the analysis described in Section 2.5, has examined both the clinical associations of these states and the reliability of their temporal properties (Khanna, Pascual-Leone, Michel et al., 2015).

Clinical findings vary across disorders. A meta-analysis of schizophrenia reported increased occurrence of class C together with reduced duration and coverage of class D (Rieger et al., 2016). In mood and anxiety disorders, reported effects were smaller and more heterogeneous, including changes in the occurrence of classes B and D (Chivu et al., 2023). These results indicate that microstates capture disease-related group differences, but they do not support interpreting individual microstate classes as diagnostic labels.

The reliability literature also shows that microstate results depend on how the templates are constructed. Using common maps across recordings produced more reliable estimates of duration, occurrence, and coverage than deriving an independent set of maps for every recording (Khanna, Pascual-Leone and Farzan, 2014). A larger test-retest study similarly found stronger reliability for duration, occurrence, and coverage than for transition measures (Kleinert et al., 2023). Cross-study comparison is further complicated because the same class letter can refer to different scalp topographies; quantitative comparison of the maps is therefore more reliable than assuming that labels such as A, B, C, and D are identical across studies (Koenig et al., 2023).

These findings make microstates relevant to representation learning as a compact description of recurring whole scalp organisation. At the same time, their dependence on template estimation and their variable clinical associations mean that they should be treated as structured neurophysiological information rather than fixed biological categories.

3.6 Evaluation under Participant and Dataset Shift

Evaluation practices are particularly important in EEG because one participant usually contributes many correlated recordings or windows. If data from the same participant appear in both training and test sets, performance can reflect stable participant characteristics rather than generalisation to unseen individuals. This issue was demonstrated in a clinical EEG archive where preventing repeated recordings from the same patients from crossing split boundaries reduced performance (Kim et al., 2023). Participant level validation, including leave-one-subject-out evaluation, therefore provides a stricter test of clinical generalisation (Miltiadous, Gionanidis et al., 2023).

For pretrained encoders, linear probing and finetuning also answer different questions. Frozen evaluation measures how directly task relevant information is available in the pretrained representation, whereas finetuning allows the encoder itself to adapt to the labelled task. Prior self-supervised EEG studies have used both approaches, and reviews report that finetuning can substantially outperform frozen evaluation (Banville et al., 2020; Kuruppu et al., 2025). Dataset shift introduces a further challenge because EEG can vary in montage, hardware, sampling rate, filtering, reference, and noise characteristics. Benchmarking and robustness studies show that such changes can reduce both representation quality and predictive performance (Engemann et al., 2021; Wagh, Wei et al., 2022). Cross-dataset evaluation is therefore needed when the intended claim extends beyond one acquisition setting.

3.7 Research Gap

Prior work shows that handcrafted EEG features and supervised neural networks remain competitive baselines, while self-supervised pretraining can learn useful EEG representations (Gemein et al., 2020; Banville et al., 2020; Kostas et al., 2021). Recent pretrained models further demonstrate different approaches to masking, channel identity, temporal context, and electrode geometry (Jiang et al., 2024; G. Wang et al., 2024; J. Wang et al., 2025; El Ouahidi et al., 2026). However, evaluation varies substantially across tasks and protocols, and it remains unclear whether self-supervised representations provide consistent benefits for resting-state clinical EEG, particularly under participant-independent and cross-dataset evaluation (Kuruppu et al., 2025).

Two related gaps concern representation design. First, coordinate-based spatial representations have been explored, but it remains unclear when explicit electrode geometry provides an advantage over learned channel identity (El Ouahidi et al., 2026). Second, microstates capture recurring whole scalp organisation and show reproducible group level associations, but they have mainly been analysed as derived EEG features rather than incorporated into self-supervised representation learning (Khanna, Pascual-Leone and Farzan, 2014; Koenig et al., 2023). These gaps motivate a controlled comparison of self-supervised representations, spatial encoding, and microstate-informed pretraining, while separately evaluating participant level and cross-dataset generalisation.

4 Methodology

This chapter describes the methods used to develop, compare, and evaluate models for resting-state EEG analysis. It is organised into seven sections covering the datasets, EEG preprocessing, the proposed models, the comparison baselines, self-supervised pretraining, downstream evaluation, and ablation studies and model analysis. Figure 4.1 provides an overview of the complete methodological workflow.

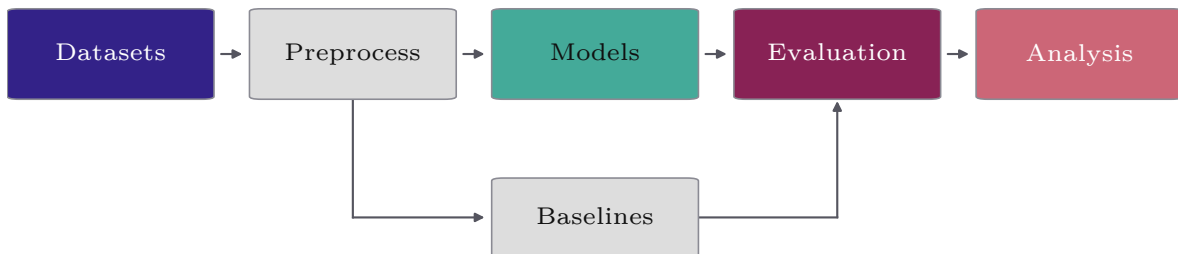


Figure 4.1: Overview of the methodological pipeline: a single preprocessing stage feeds both the self-supervised models and the comparison baselines, which are then assessed under identical evaluation protocols.

4.1 Datasets

Seven publicly available resting-state EEG datasets were included in this study. Resting-state EEG is recorded while participants are not engaged in an explicit cognitive task, typically under eyes-open or eyes-closed conditions. The selected datasets differed in cohort composition, acquisition protocol, electrode configuration, sampling rate, and recording duration. Including datasets acquired from heterogeneous cohorts and under different recording conditions was intended to reduce dependence on the characteristics of any single dataset and to support the evaluation of cross-dataset generalisation. The main characteristics of the seven datasets are summarised in Table 4.1.

Three datasets comprised healthy control participants only: the Dortmund Vital Study (DVS; Wascher et al., 2025), the SRM dataset (Hatlestad-Hall et al., 2022), and the Leipzig Study for Mind-Body-Emotion Interactions (LEMON; Babayan et al., 2019).

Table 4.1: Acquisition characteristics of the seven resting-state EEG datasets.

Dataset	Accession	Stage	Participant groups	Ch.	Frequency (Hz)	Cond.	Dur. (min)
DVS	ds005385	Pretraining	542 HC	64	1000	EO, EC	3.1
SRM	ds003775	Pretraining	111 HC	64	1024	EC	4.0
LEMON	–	Pretraining	215 HC	61	2500	EO	17.4
ADFTD	ds004504	Downstream	36 AD, 23 FTD, 29 HC	19	500	EC	13.4
FEP	ds003944	Downstream	47 FEP, 30 HC	61	1000	EO	5.1
MDD	ds003478	Downstream	32 MDD, 90 HC	64	500	EO, EC	5.8
SCZ	–	External test	30 SCZ, 47 HC	64	1000	EC	2.2

Participant groups are the participants retained after preprocessing, before the pretraining-reserve split (Figure 4.2). Ch.: EEG channels. Cond.: recording condition, eyes open (EO) or eyes closed (EC). Dur.: mean recording length. HC: healthy control. AD: Alzheimer’s disease. FTD: frontotemporal dementia. FEP: first-episode psychosis. MDD: major depressive disorder. SCZ: schizophrenia.

The remaining four datasets included both participants with a clinical diagnosis and healthy controls: the Alzheimer’s disease and frontotemporal dementia dataset (ADFTD; Miltiadous, Tzimourta et al., 2023), the first-episode psychosis dataset (FEP; Salisbury et al., 2022), the major depressive disorder dataset (MDD; Cavanagh, 2021), and the schizophrenia dataset (SCZ; Racz et al., 2025).

The datasets were allocated across three stages of the study: self-supervised pretraining, downstream clinical evaluation, and external testing. An overview of the dataset and participant allocation is provided in Figure 4.2.

DVS, SRM, and LEMON formed the primary self-supervised pretraining corpus. These datasets were selected to expose the model to resting-state EEG recordings obtained from multiple cohorts and acquisition settings. This selection was intended to reduce the likelihood that the model would learn representations specific to a single dataset. Within each dataset, 90% of the participants were allocated to the training set during self-supervised pretraining, while the remaining 10% formed a held-out validation set. The validation participants were used exclusively to monitor model performance during pretraining.

ADFTD, FEP, and MDD were used primarily for downstream clinical evaluation. However, restricting self-supervised pretraining to healthy control datasets would have limited the participant level and neurophysiological variability represented in the pretraining corpus. Therefore, 20% of participants from each of these datasets were allocated to a separate pretraining reserve subset. Each reserve subset included participants from both the clinical and healthy control groups. These subsets were added to the self-supervised pretraining corpus to broaden the range of resting-state EEG characteristics represented during pretraining. The remaining 80% of participants in each dataset formed the downstream evaluation pool.

The resulting pretraining corpus and its demographic composition are shown in Fig-

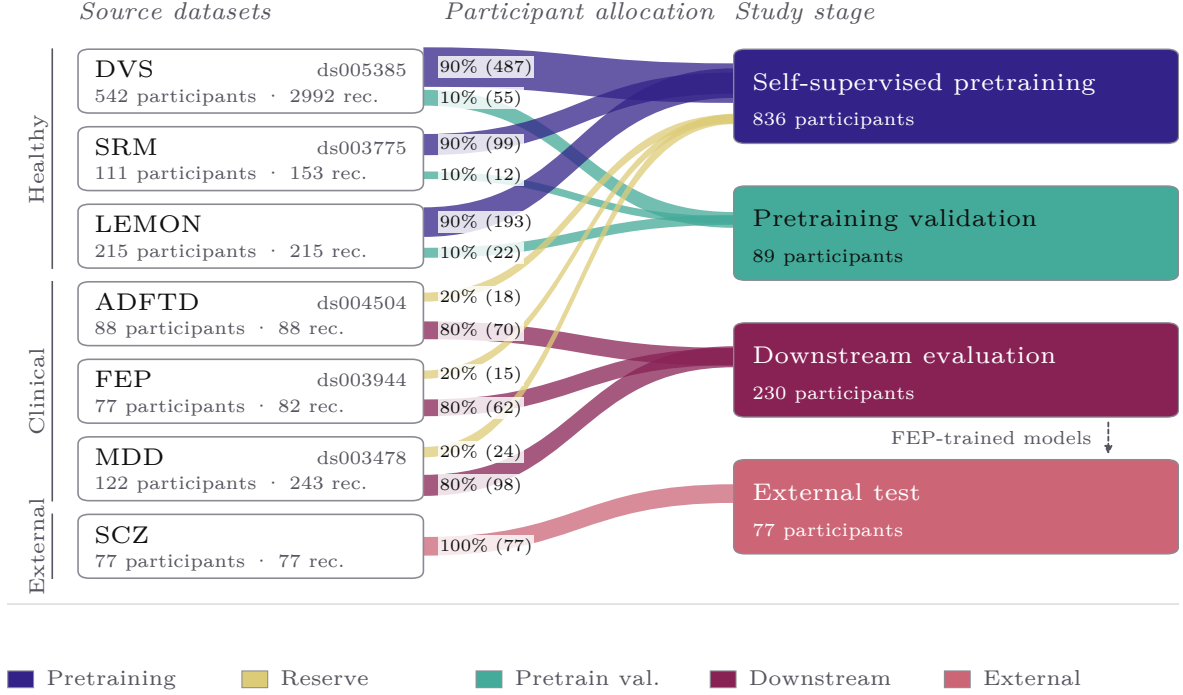


Figure 4.2: Allocation of the seven resting-state datasets and their participants to the stages of the study; partitions are defined at the participant level, so no participant contributes to both pretraining and downstream evaluation.

ure 4.3.

All dataset partitions were defined at the participant level to prevent data leakage between study stages. Participants assigned to the pretraining reserve subsets were used exclusively for self-supervised pretraining and were excluded from all downstream analyses. Consequently, no participant contributed data to both pretraining and downstream performance evaluation. This separation maintained participant level independence between the two stages.

The SCZ dataset was excluded from both self-supervised pretraining and in-domain downstream model development. Instead, it was used exclusively as an external test set for models developed using the FEP dataset. Differences between the FEP and SCZ datasets in cohort composition and acquisition conditions introduced a distribution shift between the development and external test data. This design was intended to provide a more stringent assessment of cross-dataset generalisation.

The composition of the four downstream cohorts and of the external test set, including their age and sex distributions, is shown in Figure 4.4. The cohorts differ substantially in age: the dementia cohorts are elderly, whereas the FEP and MDD cohorts are drawn from

young adults. The proportion of patients also varies across cohorts, which determines the chance level of the metric reported in Section 4.6.

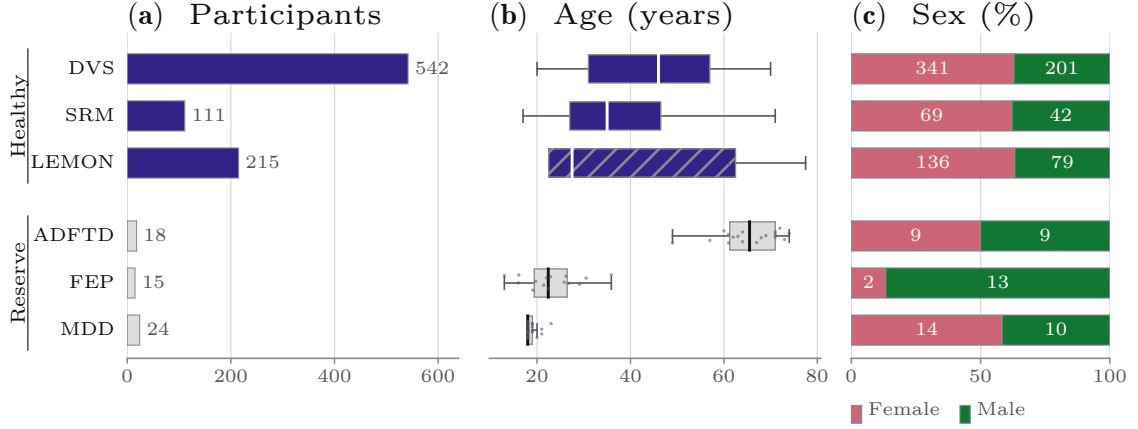


Figure 4.3: Composition of the self-supervised pretraining corpus by source dataset: participants (a), age (b), sex (c). LEMON’s age (hatched) is approximate, built from published five-year brackets.

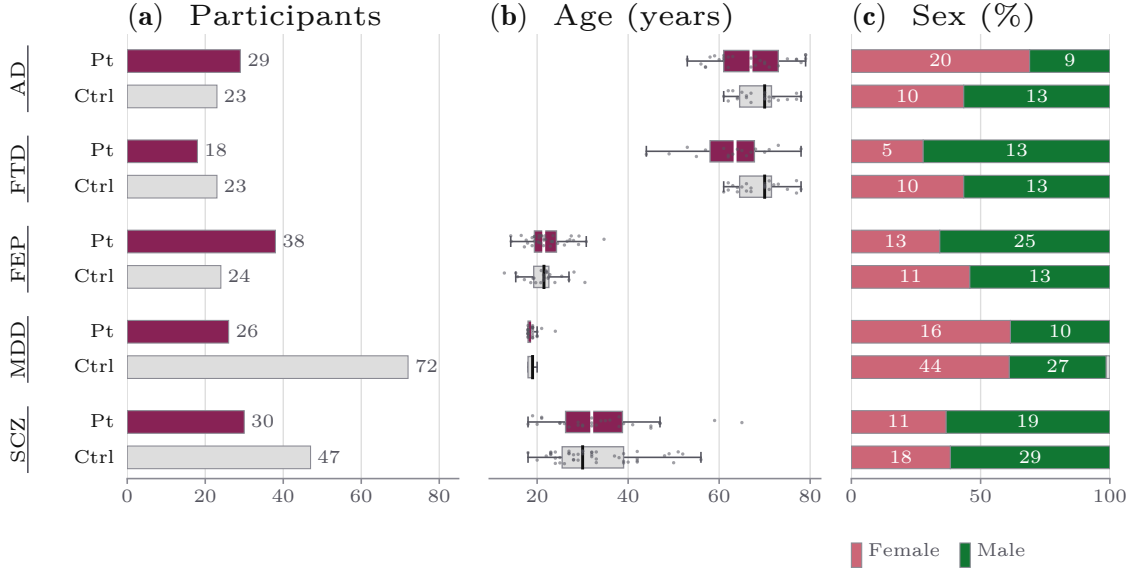


Figure 4.4: Composition of the four downstream evaluation cohorts and external SCZ test set: participants (a), age (b), and sex (c). Pt/Ctrl: patients/controls. AD and FTD share controls from the same 70-participant ADFTD pool. One MDD control had unrecorded age and sex, resulting in one fewer count in (b) and (c).

4.2 Preprocessing

EEG preprocessing was performed to reduce technical noise and physiological artefacts while preserving neural activity relevant to the subsequent analyses. Scalp-recorded EEG does not provide a noise-free representation of brain activity, as recordings may contain slow signal drift, power-line interference, malfunctioning electrodes, and artefacts caused by eye movements, cardiac activity, and muscle contractions. These sources can obscure the underlying neural signals and reduce the reliability of subsequent analyses. Because the seven datasets were collected using different acquisition systems and sampling frequencies, the same general preprocessing pipeline was applied to all datasets to improve signal comparability and produce consistently structured inputs. The complete ordered pipeline is shown in Figure 4.5.

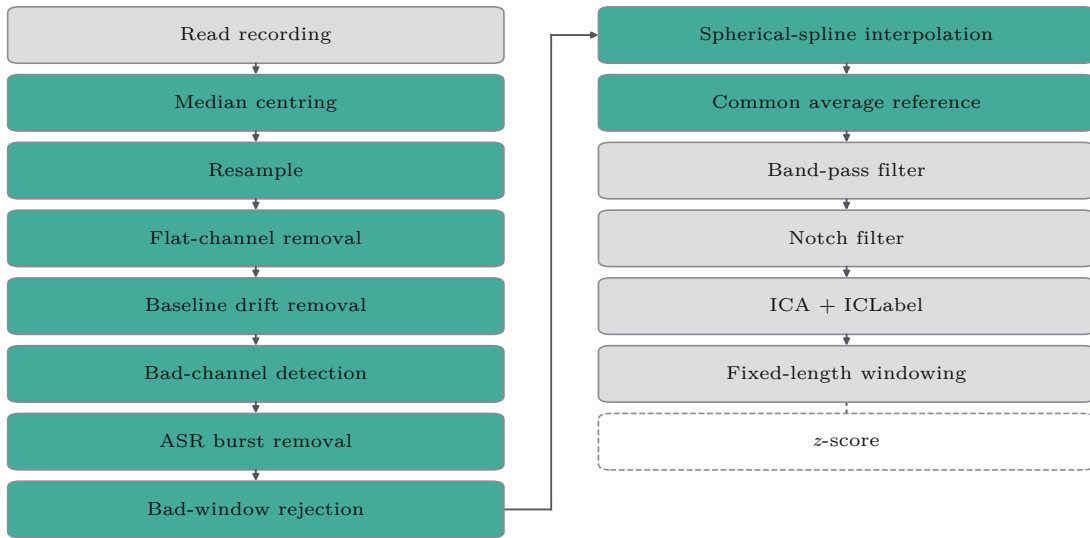


Figure 4.5: The preprocessing pipeline, shown in the order of application. Tinted stages are performed by the EEGPrep pipeline.

Each recording was loaded using the `MNE-Python` library (Gramfort, 2013). Only channels identified as EEG channels were retained, while non-EEG channels, including stimulation and auxiliary channels, were excluded. `MNE-Python` internally represents EEG amplitudes in volts; therefore, the signals were converted to microvolts by multiplying them by 10^6 . This operation changed only the unit of measurement and made the signal amplitudes easier to interpret and use during subsequent processing.

The recordings were then processed using the EEGPrep pipeline provided by the `Braindecode` library (Mullen et al., 2015; Aristimunha et al., 2026). EEGPrep applies a standardised sequence of procedures that includes offset correction, resampling,

flat channel removal, drift attenuation, bad channel detection, transient artefact correction, bad window rejection, channel interpolation, and re-referencing. Applying the same ordered procedure to all datasets reduced variability arising from dataset specific preprocessing choices.

First, the channel-wise DC offset was removed by subtracting the temporal median of each channel:

$$X'_{c,t} = X_{c,t} - \underset{\tau}{\text{median}}(X_{c,\tau}) \quad (4.1)$$

where c denotes the EEG channel, t denotes the current time sample, and τ indexes the time samples used to calculate the channel median. Throughout this section, X denotes the signal at the input of the operation being described and X' the signal it produces. Median centring removes constant channel offsets while being less sensitive than mean subtraction to isolated high-amplitude values.

The datasets were originally recorded at different sampling frequencies. All recordings were therefore resampled to a common sampling frequency of 200 Hz. Standardising the sampling frequency ensured that signals of equal duration contained the same number of samples across datasets and reduced the computational and storage requirements of the subsequent analyses. At 200 Hz, the Nyquist frequency is 100 Hz, which is substantially higher than the final upper passband frequency of 45 Hz. The resampling procedure thus retained the complete frequency range examined in this study.

Channels that remained flat for more than 5 s were identified as unusable and temporarily removed. Flat channels contain little or no electrophysiological information and may adversely affect spatial interpolation and common-average re-referencing. EEGPrep also applied its default drift removal high-pass transition band of 0.25-0.75 Hz before identifying additional unreliable channels.

Additional bad channels were detected using correlation-based procedures derived from the PREP framework (Bigdely-Shamlo et al., 2015). The signal recorded at each channel was compared with a spatial prediction derived from the remaining channels. Channels with a correlation below 0.75 were classified as unreliable. High-frequency noise measures were also used to identify electrodes dominated by non-physiological noise, even when the channels were not completely flat. The correlation threshold of 0.75 provided a relatively permissive criterion that reduced the likelihood of rejecting channels that retained usable information.

Transient high-amplitude artefacts were subsequently corrected using Artifact Subspace Reconstruction (ASR). ASR compares short-term multichannel activity with a reference distribution obtained from relatively clean portions of the recording. Signal subspaces

exhibiting abnormally high variance are identified and reconstructed using the retained subspaces of the calibration model. An ASR burst removal cutoff of 20 was used. Because higher cutoff values result in less aggressive correction, this value provided a conservative balance between suppressing large transient artefacts and preserving potentially meaningful neural activity.

After ASR, EEGPrep removed time intervals containing widespread residual contamination. A time interval was classified as unreliable when more than 25% of its channels exceeded the permitted amplitude tolerance. The default upper tolerance of seven standard deviations was used, with no finite lower threshold.

Channels previously identified as flat or unreliable were then reconstructed using spherical-spline interpolation based on the spatial distribution of the retained electrodes. Reinterpolation restored the expected channel configuration and prevented the number of input channels from varying according to the channels rejected in individual recordings.

The signals were subsequently re-referenced using a common average reference. At each time point, the mean amplitude across all retained and reconstructed EEG channels was subtracted from the amplitude of each channel:

$$X'_{c,t} = X_{c,t} - \frac{1}{C} \sum_{i=1}^C X_{i,t} \quad (4.2)$$

where C denotes the total number of EEG channels included in the reference, c denotes the channel being re-referenced, and i is the summation index. Re-referencing was performed after bad channel correction to prevent severely contaminated channels from disproportionately influencing the average reference. The common average reference also provided a consistent reference convention across datasets that may originally have been recorded using different physical references. However, its effectiveness depends on the number and spatial coverage of the available electrodes.

Following EEGPrep, the recordings were band-pass filtered between 0.5 and 45 Hz using the default zero-phase finite impulse response filtering procedure provided by **MNE-Python**. The lower cutoff attenuated slow baseline drift, while the upper cutoff reduced high-frequency instrumental and muscular noise. Activity outside this frequency range was excluded from the present analysis rather than assumed to be entirely non-neural.

A notch filter was then applied at the local electrical mains frequency. A frequency of 50 Hz was used for the European datasets, while 60 Hz was used for the datasets recorded in the United States. The notch filter was applied as a fixed component of the standardised preprocessing pipeline. Because the preceding 45 Hz low-pass filter already attenuated activity at 50 and 60 Hz, the notch filter was expected to provide only

limited additional suppression of power-line interference. It was nevertheless retained to maintain a uniform procedure across datasets.

Independent component analysis (ICA; Hyvärinen and Oja, 2000) was subsequently used to reduce structured physiological and non-neural artefacts that remained after the preceding cleaning operations. For each recording, a separate copy of the signal was high-pass filtered at 1 Hz. FastICA was then fitted to this copy using a maximum of 20 independent components. For recordings containing fewer channels, the number of components was limited to one fewer than the number of available EEG channels. A fixed random seed of 42 was used to improve the reproducibility of the decomposition. Fitting ICA on a 1 Hz high-pass-filtered copy reduced the influence of slow signal fluctuations and improved the stability of the decomposition.

The resulting independent components were automatically classified using ICLabel through the MNE-ICLabel library (Li et al., 2022). ICLabel estimates the probability that each component represents brain activity, muscle artefact, ocular artefact, cardiac activity, line noise, channel noise, or other activity. A component was excluded when its predicted class was muscle artefact, eye blink artefact, cardiac activity, line noise, or channel noise and the corresponding classification probability was at least 0.80. This relatively high threshold was used as a conservative criterion so that components were removed only when the classifier expressed strong confidence that they were artefactual.

The ICA solution estimated from the 1 Hz high-pass-filtered copy was then applied to the corresponding 0.5-45 Hz signal. This procedure allowed the decomposition to be estimated from data with reduced slow drift while preserving the lower-frequency activity included in the final analysis.

The effect of the complete pipeline on a single recording is shown in Figure 4.6, together with the independent components that ICLabel classified for that recording.

One limitation of this procedure is that ICLabel was originally developed using extended-Infomax ICA decompositions, whereas the present pipeline used FastICA. The data otherwise followed the principal ICLabel preprocessing recommendations because they were common-average referenced and ICA was fitted using a 1 Hz high-pass-filtered copy. Nevertheless, the difference in ICA algorithm may affect the reliability of the automatic component classifications and should be considered when interpreting the results.

Following artefact removal, each recording was divided into fixed-length, non-overlapping windows of 15 s. At a sampling frequency of 200 Hz, each window contained 3000 time samples. A fixed window duration ensured that all inputs had identical temporal dimensions. The 15 s duration provided a compromise between temporal resolution, computational requirements, the number of available windows, and the amount of temporal context represented within each window. In particular, a 15 s window contains approximately 7.5 cycles of activity at the lower passband frequency of 0.5 Hz.

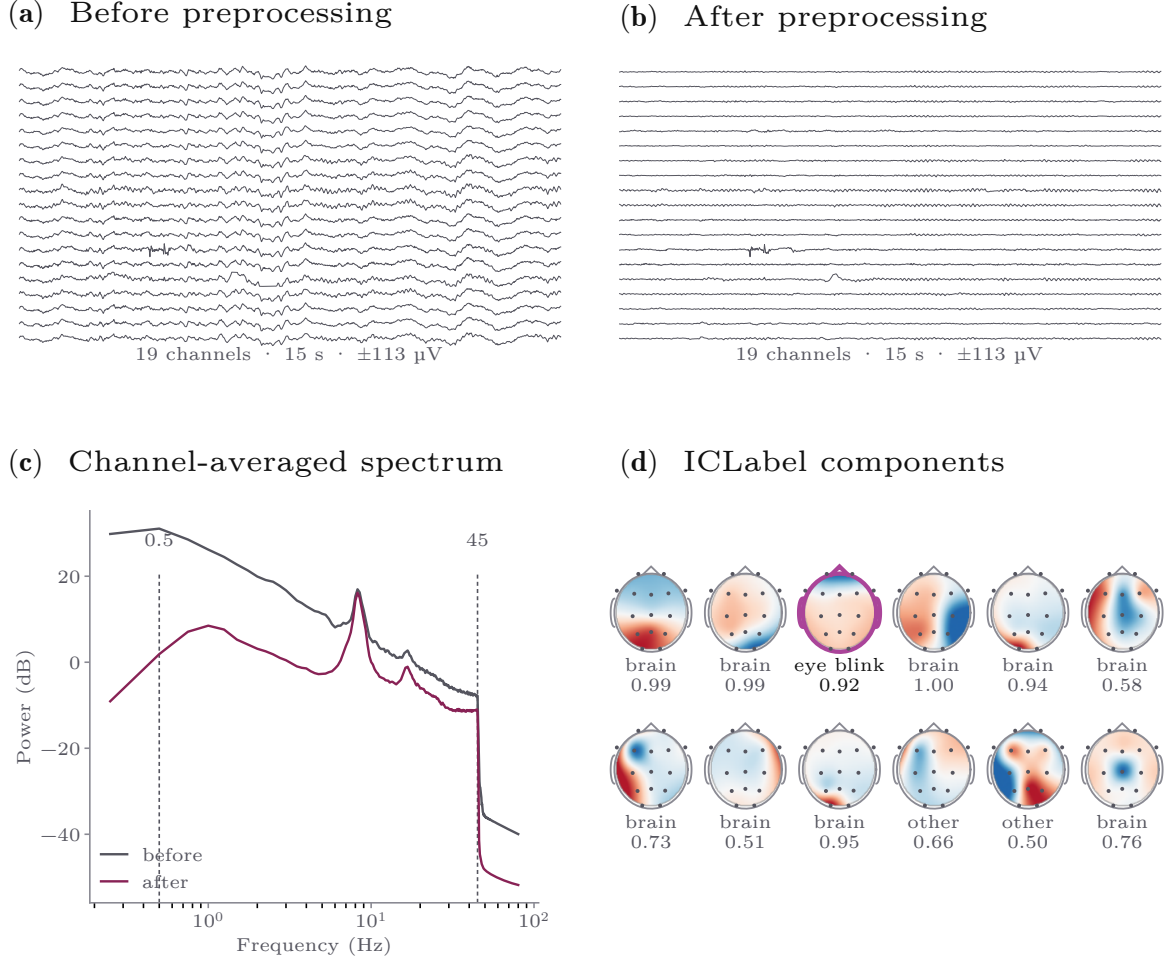


Figure 4.6: Preprocessing pipeline applied to a healthy subject (sub-037) from ADFTD dataset. **(a,b)** The same 15-s excerpt before and after preprocessing, shown at a shared amplitude scale defined by the 99.5th percentile of the raw signal amplitude, so that the removal of slow drift and large transients is directly visible. **(c)** Channel-averaged power spectral density before and after preprocessing, with the passband edges marked. **(d)** Independent components with their ICLabel class and probability; outlined components were removed based on an artefact-class probability of ≥ 0.80 .

Non-overlapping windows were used to avoid generating highly similar samples containing duplicated time points. Any incomplete window containing fewer than 3000 samples at the end of a recording was discarded.

Z-score normalisation was not applied as part of the preprocessing pipeline described above and was not stored in the saved EEG windows. Instead, it was performed when the data were loaded for the supervised deep learning and self-supervised learning models. Each channel within each 15 s window was normalised independently using the temporal mean and standard deviation calculated from that channel within the same window:

$$\tilde{X}_{w,c,t} = \frac{X_{w,c,t} - \mu_{w,c}}{\sigma_{w,c} + \varepsilon} \quad (4.3)$$

where w denotes the window, c denotes the EEG channel, t denotes the time sample, and ε is a small positive constant included to prevent division by zero. The within-window temporal mean was calculated as

$$\mu_{w,c} = \frac{1}{T} \sum_{\tau=1}^T X_{w,c,\tau} \quad (4.4)$$

and the corresponding standard deviation was calculated as

$$\sigma_{w,c} = \sqrt{\frac{1}{T} \sum_{\tau=1}^T (X_{w,c,\tau} - \mu_{w,c})^2} \quad (4.5)$$

where $T = 3000$ denotes the number of time samples in each window and τ is the summation index over those samples. Because the mean and standard deviation were calculated independently for each channel and window, no normalisation statistics were shared across participants or between the training, validation, and test partitions. This procedure therefore did not introduce information leakage between the data partitions.

The traditional machine learning models followed a different normalisation procedure. The EEG windows used for these models were not z-score normalised during data loading. Instead, the extracted feature vectors were standardised using a standard scaler fitted exclusively to the training fold and subsequently applied to the corresponding validation and test folds. This ensured that the traditional machine learning analyses also avoided information leakage.

4.3 Proposed Models

This study developed three self-supervised models for learning representations from resting-state EEG: BrainLM-EEG, BrainLM-EEG-RoPE, and BrainLM-EEG-Microstate. BrainLM (Caro et al., 2024) was originally introduced as a foundation model for fMRI and uses masked autoencoding (He et al., 2022) to learn spatiotemporal representations from brain activity. Here, the BrainLM framework was adapted to EEG, resulting in BrainLM-EEG, which serves as the reference architecture for the two subsequent extensions.

Adapting BrainLM to EEG required reformulating its spatiotemporal input representation around EEG electrodes and short temporal signal patches. BrainLM-EEG retains the core masked encoder-decoder learning principle while representing each EEG window as a sequence of channel-specific temporal patches. Building on this architecture, BrainLM-EEG-RoPE replaces learned spatial channel embeddings with geometry-aware rotary positional encoding derived from electrode locations, while BrainLM-EEG-Microstate augments the masked reconstruction objective with an auxiliary prediction task based on canonical EEG microstates.

Each extension modifies a single aspect of BrainLM-EEG while retaining the same input representation, Transformer backbone, masking strategy, and pretraining data. This controlled design enables the effect of each modification to be evaluated separately.

The proposed models were compared with three groups of established methods, described in Section 4.4.

4.3.1 BrainLM-EEG: Adapting BrainLM to EEG

BrainLM-EEG adapted the masked autoencoding framework of BrainLM from fMRI to resting-state EEG. The original BrainLM represents fMRI as spatiotemporal patches defined over brain parcels and time. In BrainLM-EEG, the corresponding spatial units are EEG electrodes, and each electrode signal is divided into short temporal patches. The model is pretrained by reconstructing randomly masked channel-time patches from the remaining visible EEG context. Unless stated otherwise, the architectural and training choices described in this subsection were shared by BrainLM-EEG, BrainLM-EEG-RoPE, and BrainLM-EEG-Microstate.

Input representation and tokenisation

Following preprocessing, each EEG recording was segmented into non-overlapping 15-second windows. Because the included datasets differed in electrode configuration, the

analysis was restricted to a common set of 19 channels corresponding to the standard international 10-20 montage. The electrodes and the order in which the model received them are listed in Appendix A.2. All signals were resampled to 200 Hz. Each input window was therefore represented as

$$\mathbf{X} \in \mathbb{R}^{C \times T}, \quad C = 19, \quad T = 3000 \quad (4.6)$$

where C denotes the number of channels and T denotes the number of time points.

Each channel was divided into 15 non-overlapping temporal patches of one second, with 200 time points per patch. A 15-second window consequently produced

$$N = C \times P = 19 \times 15 = 285 \quad (4.7)$$

spatiotemporal tokens, where $P = 15$ denotes the number of temporal patches per channel. Each token represented one electrode during a specific one-second interval. A linear projection mapped each 200-sample patch to a 512-dimensional embedding. The construction of the token sequence is shown in Figure 4.7.

Positional representation and sequence construction

The reference BrainLM-EEG model used separate learnable embeddings for spatial and temporal position. For each token, the patch embedding was combined with a channel embedding identifying the corresponding electrode and a temporal embedding identifying its position within the 15-second window.

A learnable classification token, denoted CLS, was prepended to the token sequence. The CLS token had the same 512-dimensional embedding size as the patch tokens and was intended to aggregate information across the complete input window. Its final encoder representation was used as the global representation for downstream classification.

Encoder-decoder architecture

The encoder comprised four transformer blocks with an embedding dimension of 512 and four self-attention heads. The dimension of each attention head was therefore 128. Within each block, the feed-forward network expanded the representation dimension by a factor of four before projecting it back to 512 dimensions. Gaussian error linear unit (GELU; Hendrycks and Gimpel, 2023) activations were used in the feed-forward layers.

The decoder comprised two transformer blocks with the same embedding dimension and number of attention heads. It was deliberately shallower than the encoder so that the encoder was required to learn informative latent representations that supported reconstruction using a comparatively lightweight decoder. A linear reconstruction head mapped each decoder output to the 200 signal values of the corresponding one-second patch.

Masked reconstruction objective

During self-supervised pretraining, 75% of the 285 spatiotemporal tokens were selected at random and masked. Only the visible patch tokens and the CLS token were passed to the encoder. Learnable mask tokens were then inserted at the masked positions before the complete sequence was passed to the decoder.

The decoder predicted the original values of the masked patches. The reconstruction loss was the mean squared error (MSE) calculated only at masked token positions:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2 \quad (4.8)$$

where $\mathcal{M}$ denotes the set of masked tokens, $\mathbf{x}_i$ denotes the original signal patch, and $\hat{\mathbf{x}}_i$ denotes its reconstruction.

This pretrained EEG model formed the reference against which the two additional BrainLM-EEG variants were assessed. The shared architecture and the two variants are summarised in Figure 4.7.

4.3.2 Extensions to BrainLM-EEG

The two additional variants addressed complementary limitations of the BrainLM-EEG reference architecture. BrainLM-EEG-RoPE introduced an explicit representation of relative electrode geometry, whereas BrainLM-EEG-Microstate incorporated a neurophysiologically informed target into the self-supervised learning objective. In both variants, all components not directly related to the proposed modification were kept unchanged. The comparison with BrainLM-EEG therefore isolates the effect of each extension from the broader contribution of adapting BrainLM to EEG.

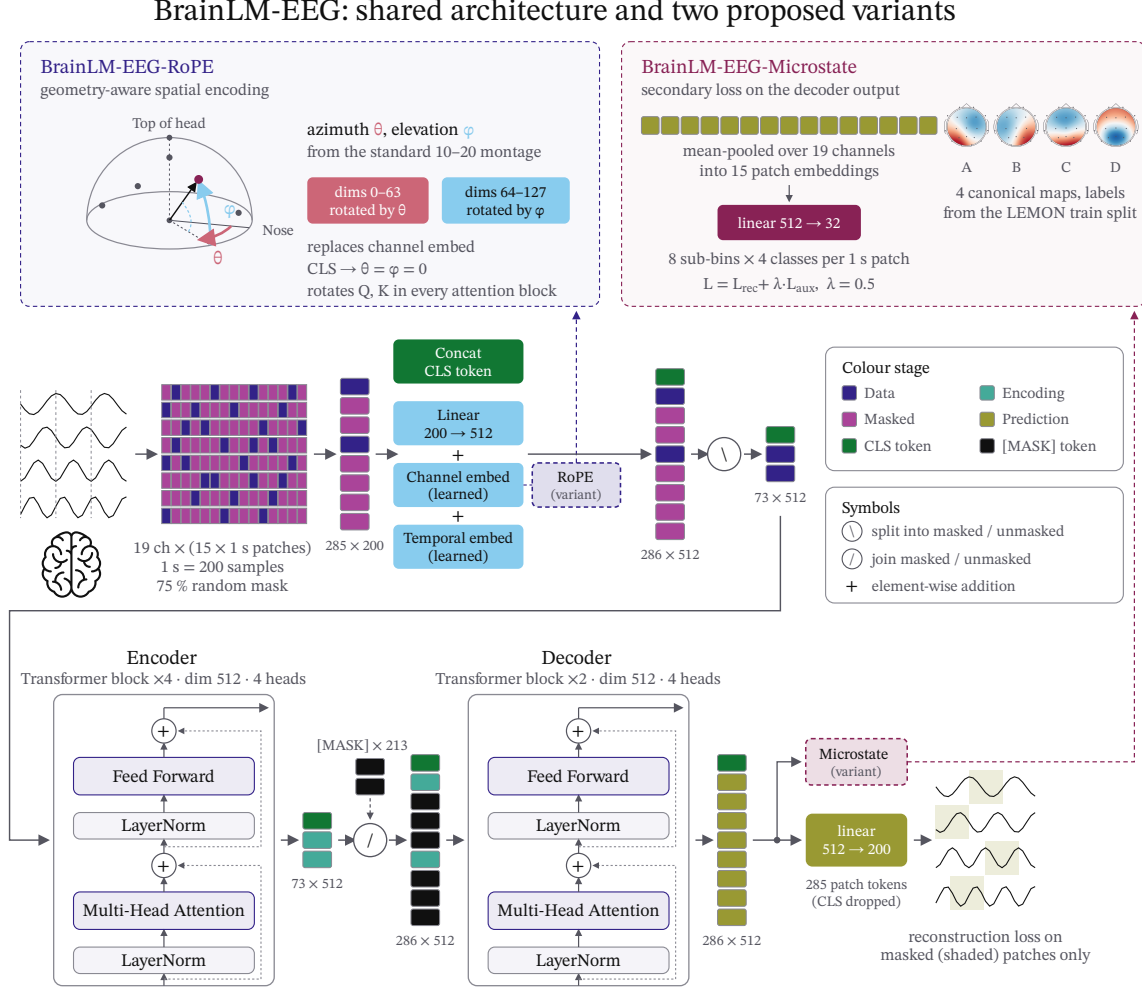


Figure 4.7: Overview of the shared BrainLM-EEG architecture and its two variants. The backbone performs masked reconstruction of multichannel EEG using channel-time patch embeddings, a transformer encoder-decoder, and a linear reconstruction head. BrainLM-EEG-RoPE replaces channel embeddings with electrode specific rotary positional encoding based on azimuth and elevation. BrainLM-EEG-Microstate adds an auxiliary classification head that predicts the microstate class of each 125 ms sub-interval.

Geometry-Aware Variant: BrainLM-EEG-RoPE

Self-attention does not inherently encode token position. In BrainLM-EEG, spatial position was represented by a learned embedding for each EEG channel. Although this allowed the model to distinguish electrodes, it did not explicitly represent the geometric relationships between them. EEG electrodes are distributed over the approximately spherical surface of the scalp rather than along a one-dimensional sequence. Each electrode c was therefore represented by an azimuth angle θ_c and an elevation angle φ_c , obtained from the standard 10-20 montage coordinates. The coordinate convention, the conversion to these two angles, and their values for each electrode are given in Appendix A.2.

BrainLM-EEG-RoPE replaced the learned spatial channel embedding with a rotary positional encoding based on these electrode coordinates. The learnable temporal embedding was retained, so the proposed modification affected only the representation of spatial position.

Rotary positional encoding (RoPE; Su et al., 2024) applies position-dependent rotations to the query and key vectors within self-attention. Consider one attention head with query vector $\mathbf{q} \in \mathbb{R}^d$ and key vector $\mathbf{k} \in \mathbb{R}^d$. Each vector is partitioned into $d/2$ consecutive two-dimensional coordinate pairs. For a token at scalar position m , pair i is rotated by the angle $m\omega_i$, giving the rotation block

$$\mathbf{R}_i(m) = \begin{bmatrix} \cos(m\omega_i) & -\sin(m\omega_i) \\ \sin(m\omega_i) & \cos(m\omega_i) \end{bmatrix}, \quad \omega_i = b^{-2i/d}, \quad i = 0, \dots, \frac{d}{2} - 1 \quad (4.9)$$

where $b = 10^4$. Because rotation matrices satisfy $\mathbf{R}(m)^\top \mathbf{R}(n) = \mathbf{R}(n - m)$, the attention score between a query at position m and a key at position n can be written as

$$(\mathbf{R}(m)\mathbf{q})^\top (\mathbf{R}(n)\mathbf{k}) = \mathbf{q}^\top \mathbf{R}(n - m)\mathbf{k} \quad (4.10)$$

The resulting attention score therefore depends on the relative displacement between positions rather than only on their absolute locations.

Two-dimensional RoPE has previously been used by Heo et al. (2025) to represent horizontal and vertical patch coordinates in vision transformers. BrainLM-EEG-RoPE adapted this principle to EEG electrode geometry. Within each attention head, the first half of the query and key dimensions were associated with electrode azimuth, while the second half were associated with electrode elevation. This allocation yielded $d/4$ rotation pairs for each coordinate. The rotations were applied to the query and key vectors in every self-attention layer immediately before the dot-product attention calculation.

For tokens associated with electrodes i and j , the resulting attention score satisfied

$$(\mathbf{R}_{2D}(\theta_i, \varphi_i)\mathbf{q})^\top (\mathbf{R}_{2D}(\theta_j, \varphi_j)\mathbf{k}) = \mathbf{q}^\top \mathbf{R}_{2D}(\theta_j - \theta_i, \varphi_j - \varphi_i)\mathbf{k} \quad (4.11)$$

Thus, each attention score incorporated the relative azimuthal and elevational displacement between the corresponding electrodes. The CLS token had no scalp location and was assigned $\theta = 0$ and $\varphi = 0$. The rotary encoding was parameter-free with respect to channel identity and depended only on electrode coordinates, allowing the same mechanism to be applied to montages or electrode locations not observed during pretraining.

The structure that the rotary encoding imposes on inter-electrode attention, and the corresponding structure of the learned channel embedding that it replaces, are shown in Figure 4.8.

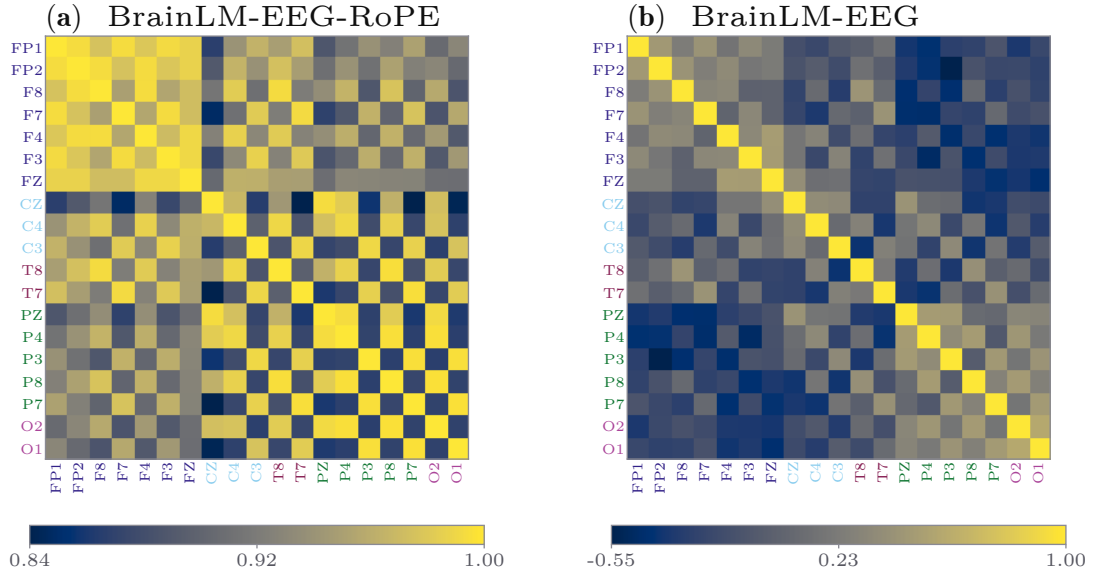


Figure 4.8: Comparison of spatial structure in BrainLM-EEG-RoPE and BrainLM-EEG. (a) Expected pairwise attention scores induced by RoPE from the electrode coordinates and fixed throughout training. (b) Pairwise cosine similarity between the channel embeddings learned by BrainLM-EEG after pretraining. Both show spatial organisation across electrodes, but the panels use different quantities and scales and should therefore be compared by pattern rather than magnitude.

A concurrent positional encoding benchmark by Yuce and Stober (2026) also represented electrode azimuth and elevation using sinusoidal functions. That approach generated an absolute positional vector that was added to each token embedding. In contrast, BrainLM-

EEG-RoPE applied a multiplicative rotation within the attention operation and encoded relative inter-electrode geometry without altering the token content representation.

Microstate-Informed Variant: BrainLM-EEG-Microstate

BrainLM-EEG was pretrained using masked reconstruction of raw EEG signal amplitudes. BrainLM-EEG-Microstate extended this objective with an auxiliary prediction task intended to encourage the model to retain information about neurophysiologically meaningful resting-state scalp topographies. The auxiliary objective was added without changing the BrainLM-EEG encoder-decoder architecture or masking procedure.

During resting-state EEG, the spatial distribution of scalp potentials remains approximately stable for brief intervals, typically around 80-120 ms, before transitioning to another configuration. These quasi-stable topographies are referred to as EEG microstates (Khanna, Pascual-Leone and Farzan, 2014) and are described in Section 2.5. Four canonical microstate classes, conventionally labelled A, B, C, and D, have been repeatedly identified in resting-state EEG and account for a substantial proportion of the global topographic variance. Their temporal characteristics have also been reported to differ across several neurological and psychiatric conditions (Khanna, Pascual-Leone and Farzan, 2014; Michel and Koenig, 2018). These properties motivated the use of microstate information as an additional learning target during BrainLM-EEG pretraining.

The canonical microstate maps used in this study were estimated once before self-supervised pretraining and were subsequently held fixed. They were derived exclusively from the training partition of the LEMON dataset, which formed part of the pretraining corpus. No participants assigned to downstream evaluation were used to estimate these maps. The resulting maps are shown in Figure 2.3.

The number of microstate prototypes was fixed to $K = 4$, corresponding to the four canonical classes A-D. Microstate topographies were clustered using the polarity-invariant modified k -means algorithm (Pascual-Marqui et al., 1995), implemented with the `Pycrostates` Python library (Férat et al., 2022). The resulting prototypes were matched to the conventional A-D ordering by comparison with the reference maps reported by Michel and Koenig (2018). The fixed prototype matrix was denoted by

$$\mathbf{M} = [\mathbf{m}_1 \quad \mathbf{m}_2 \quad \cdots \quad \mathbf{m}_K] \in \mathbb{R}^{C \times K} \quad (4.12)$$

where $C = 19$ denotes the number of EEG channels, $K = 4$ denotes the number of microstate classes, and $\mathbf{m}_k \in \mathbb{R}^C$ represents the canonical scalp topography of microstate class k .

To generate the target labels for the auxiliary task, each one-second temporal patch was divided into $S = 8$ non-overlapping subintervals of 125 ms. At the sampling frequency of 200 Hz, each subinterval contained 25 time points. Let $p \in \{1, \dots, P\}$ index the one-second temporal patches within a 15-second input window, where $P = 15$, and let $s \in \{1, \dots, S\}$ index the 125-ms subintervals within temporal patch p .

For each subinterval, the signal was averaged over its 25 time points across each of the C channels, producing a scalp topography $\mathbf{z}_{p,s} \in \mathbb{R}^C$. The observed topography and each canonical microstate map were mean-centred across channels, yielding $\hat{\mathbf{z}}_{p,s}$ and $\hat{\mathbf{m}}_k$, respectively. The target microstate label was assigned to the canonical map with the highest absolute spatial correlation:

$$y_{p,s} = \arg \max_{k \in \{1, \dots, K\}} \left| \frac{\hat{\mathbf{z}}_{p,s}^\top \hat{\mathbf{m}}_k}{\|\hat{\mathbf{z}}_{p,s}\| \|\hat{\mathbf{m}}_k\|} \right| \quad (4.13)$$

where $y_{p,s} \in \{1, \dots, K\}$ denotes the microstate class assigned to subinterval s of temporal patch p . The absolute value makes the assignment polarity-invariant, such that a scalp topography and its sign-reversed counterpart are treated as belonging to the same microstate class.

The auxiliary task required the model to predict the microstate class of each of the $S = 8$ subintervals within every one-second temporal patch. Predictions were obtained from the decoder representations. For each 15-second input window, the decoder produced a 512-dimensional representation for each of the $C \times P = 19 \times 15 = 285$ channel-time tokens.

For a given temporal patch p , the $C = 19$ decoder representations corresponding to the same one-second interval were averaged across channels. This produced a single 512-dimensional representation for that temporal patch. Averaging across channels was used because a microstate label describes the spatial configuration of potentials across the scalp at a given time, rather than the activity of an individual electrode.

A linear prediction head mapped the pooled representation of each temporal patch to $S \times K = 8 \times 4 = 32$ logits. These logits represented eight separate four-class predictions, one for each 125-ms subinterval. Let $\mathbf{e}_{p,s} \in \mathbb{R}^K$ denote the vector of K logits predicted for subinterval s of temporal patch p . The element $\mathbf{e}_{p,s}[k]$ is therefore the logit assigned by the model to microstate class k , whereas $y_{p,s}$ is the target class defined in Equation 4.13.

The auxiliary loss was the categorical cross-entropy between the predicted class distribution and the target microstate class, averaged across all P temporal patches and all S subintervals:

$$\mathcal{L}_{\text{aux}} = -\frac{1}{PS} \sum_{p=1}^P \sum_{s=1}^S \log \frac{\exp(\mathbf{e}_{p,s}[y_{p,s}])}{\sum_{k=1}^K \exp(\mathbf{e}_{p,s}[k])} \quad (4.14)$$

Here, the denominator applies the softmax normalisation across the K candidate microstate classes. The numerator contains the exponential of the logit assigned to the target class $y_{p,s}$. Thus, for each subinterval, the loss is the negative logarithm of the probability assigned by the model to its target microstate class.

The auxiliary microstate loss was computed for all temporal patches, regardless of which channel-time tokens were masked. In contrast, the reconstruction loss $\mathcal{L}_{\text{rec}}$ remained unchanged and was calculated only at masked token positions. This differs from masked prediction approaches such as Bao et al. (2022) and Baevski et al. (2022), where the prediction objective is restricted to masked positions.

Restricting the auxiliary loss to masked positions was not well suited to the tokenisation and masking scheme used here. Masking was applied independently to individual channel-time tokens, whereas a microstate label represented the scalp topography across all channels within a temporal subinterval. At the 75 % masking ratio, a given one-second temporal patch generally contained a mixture of masked and visible channel-time tokens. Moreover, the auxiliary prediction was obtained by pooling decoder representations across all channels. Consequently, there was no direct one-to-one correspondence between an individual masked token and a microstate target. The auxiliary loss was therefore evaluated for all temporal patches, independently of the reconstruction mask.

The reconstruction and auxiliary objectives were combined as

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{aux}}, \quad \lambda = 0.5 \quad (4.15)$$

where λ controls the contribution of the microstate objective relative to the reconstruction objective and was fixed to 0.5 in this study.

The auxiliary objective encouraged the model to retain information about canonical scalp topographies and their temporal progression alongside the information required for waveform reconstruction. Microstate labels were used only during pretraining, and the auxiliary prediction head was discarded afterwards. The same BrainLM-EEG encoder was then used for downstream evaluation, so any performance differences reflected the pretraining objective rather than differences in model capacity.

4.4 Comparison Baselines

Three complementary groups of comparison baselines were included: classical machine learning models trained on handcrafted features, supervised deep neural networks trained directly on raw EEG, and pretrained EEG foundation models evaluated using linear probing. Together, these baselines provided comparisons with conventional feature engineering, task specific end to end representation learning, and transfer learning from existing pretrained EEG models.

4.4.1 Classical Machine Learning Baselines

The classical baselines used handcrafted EEG features rather than raw waveforms. Features were extracted separately from each channel of every 15-second window and concatenated into a single feature vector. Four feature families were included: spectral band power, time-domain statistics, discrete-wavelet energy, and continuous-wavelet energy. These feature families followed the approach used in EEGBench (Kastrati et al., 2025) and are summarised in Table 4.2.

Spectral features were computed using Welch’s method (Welch, 1967), which estimates the power spectral density by averaging spectra from overlapping signal segments. Band power was then calculated within five EEG frequency bands: δ (0.5-4 Hz), θ (4-8 Hz), α (8-13 Hz), β (13-30 Hz), and γ (30-45 Hz).

The time-domain features comprised the mean, variance, skewness, kurtosis, root-mean-square amplitude, zero-crossing rate, Hjorth activity, Hjorth mobility, Hjorth complexity (Hjorth, 1970), line length, and peak-to-peak amplitude. Discrete-wavelet features consisted of the energy of the approximation and detail coefficients obtained using a Daubechies-4 wavelet (Daubechies, 1988). Continuous-wavelet features consisted of Morlet-wavelet energy within the same five EEG frequency bands (Torrence and Compo, 1998). Where applicable, feature magnitudes were transformed using $\log(1 + x)$. Each feature was then standardised to zero mean and unit variance using parameters estimated from the training partition only.

The resulting feature vectors were classified using linear discriminant analysis (LDA) (Fisher, 1936), a linear-kernel support vector machine (SVM) (Cortes and Vapnik, 1995), and XGBoost (T. Chen and Guestrin, 2016).

Table 4.2: Handcrafted feature families used for the classical machine learning baselines. Features were extracted separately from each channel and subsequently concatenated. Here, L denotes the number of decomposition levels used for the discrete-wavelet transform.

Feature family	Features represented	Features/channel
Spectral band power	Welch band power in the δ , θ , α , β , and γ bands	5
Time-domain statistics	Amplitude and temporal signal characteristics, including the Hjorth parameters	11
Discrete-wavelet energy	Energy of the approximation and detail coefficients from a Daubechies-4 decomposition	$L + 1$
Continuous-wavelet energy	Morlet-wavelet energy in the five EEG frequency bands	5

4.4.2 Supervised Deep Learning Baselines

The supervised deep learning baselines were trained end to end on each downstream dataset. These models received the preprocessed EEG windows directly as input, with $\mathbf{X} \in \mathbb{R}^{C \times T}$. All architectures were loaded from the **Braindecode** library (Aristimunha et al., 2026), randomly initialised, and trained using early stopping based on validation loss.

To represent different deep learning approaches to EEG modelling, three architectures were evaluated. EEGNeX (X. Chen et al., 2024) uses a compact convolutional architecture with dilated and depthwise-separable convolutions. EEGConformer (Song et al., 2023) combines convolutional feature extraction with a transformer encoder, allowing both local and longer-range temporal dependencies to be modelled. ATCNet (Altaheri et al., 2023) combines convolutional feature extraction, sliding window multi-head self-attention, and temporal convolutional modules. Together, these architectures provided a set of supervised baselines with different modelling strategies, against which the pretrained BrainLM-EEG representations could be compared.

4.4.3 Pretrained EEG Foundation Model Baselines

The final comparison group consisted of publicly available pretrained EEG foundation models. Three models were evaluated: LaBraM (Jiang et al., 2024), CBraMod (J. Wang et al., 2025), and REVE (El Ouahidi et al., 2026). For each model, the publicly released

pretrained weights were loaded and the representation backbone was frozen. A linear classification head was then trained on the corresponding downstream training partition. This linear probing protocol assessed the separability of the diagnostic classes in each pretrained representation space without modifying the encoder.

All BrainLM-EEG models and comparison baselines were evaluated using the same seed-controlled participant level partitions. Preprocessing, training, validation, and test partitions were held constant wherever permitted by the input requirements of the respective models. The same downstream metrics and model selection criteria were used across model families. This design reduced the extent to which performance differences could be attributed to differences in participant allocation or evaluation procedure.

4.5 Self-Supervised Pretraining

BrainLM-EEG, BrainLM-EEG-RoPE, and BrainLM-EEG-Microstate were pretrained using the same data partitions and optimisation procedure. The variants differed only in the architectural or objective level modifications described in Section 4.3.2. The remaining pretraining settings were kept fixed to support a direct comparison between the model variants.

The pretraining corpus comprised resting-state EEG data from six datasets, allocated as described in Section 4.1: the training partitions of DVS, SRM, and LEMON, together with the pretraining reserve subsets of ADFTD, FEP, and MDD. Because the clinical reserve subsets were comparatively small, they were used entirely for pretraining and were not divided further.

The resulting corpus and its demographic composition are shown in Figure 4.3. The three healthy control datasets contribute the large majority of the participants, while the clinical reserve subsets contribute the diagnostic and age variability that those datasets lack.

All models used the common set of 19 EEG channels described in Section 4.3.1. Recordings were divided into non-overlapping 15-second windows sampled at 200 Hz, giving inputs of $\mathbf{X} \in \mathbb{R}^{19 \times 3000}$. Per-window, per-channel z -score normalisation was applied during data loading as described in Section 4.2. The normalised windows were then processed using the spatiotemporal patching, tokenisation, masking, and reconstruction procedures described in Section 4.3.1.

All models were optimised using AdamW (Loshchilov and Hutter, 2019) with a batch size of 512. The learning rate was set using the linear scaling rule (Goyal et al., 2018), with a base learning rate of 1.5×10^{-4} defined for a batch size of 256. For the batch

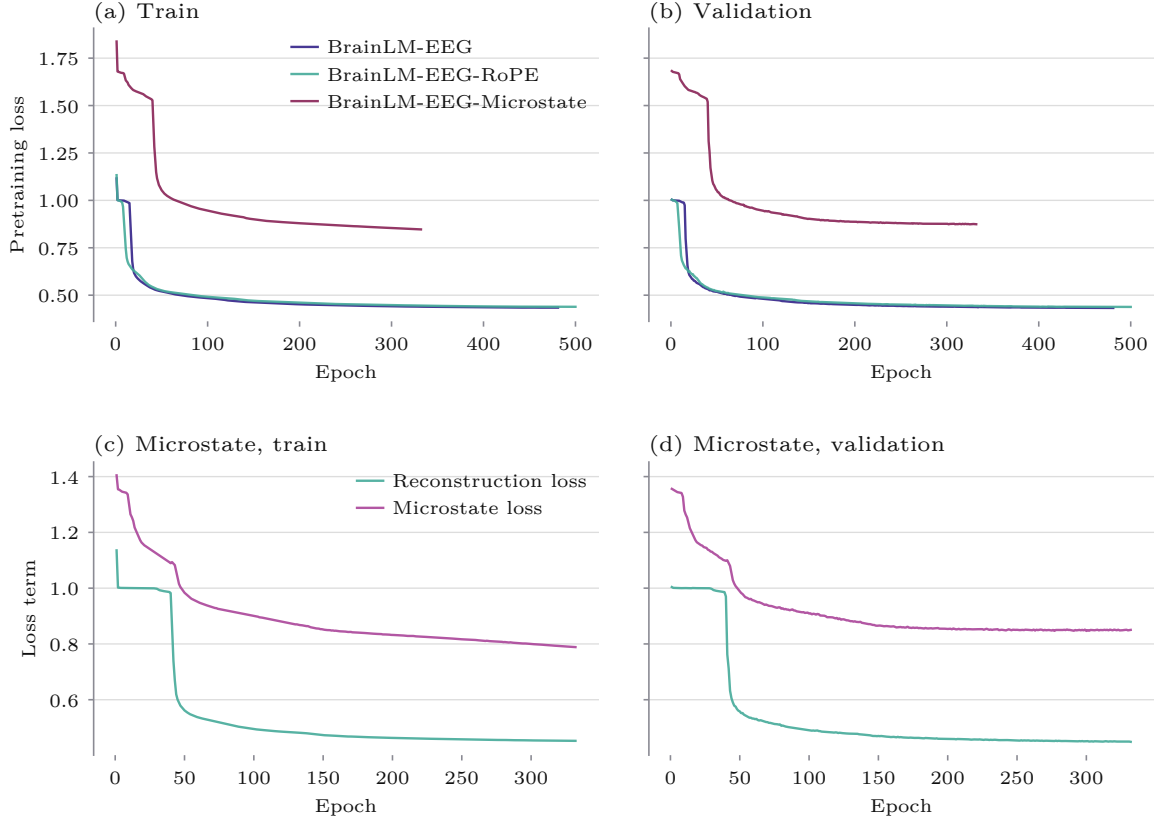


Figure 4.9: Pretraining loss trajectories for the three model variants; lines end at each variant’s early-stopped epoch. **(a,b)** Train and validation total loss. For BrainLM-EEG and BrainLM-EEG-RoPE, this is the reconstruction loss $\mathcal{L}_{\text{rec}}$; for BrainLM-EEG-Microstate, it is the combined objective $\mathcal{L} = \mathcal{L}_{\text{rec}} + 0.5\mathcal{L}_{\text{aux}}$ from Equation 4.15. **(c,d)** Separate reconstruction and microstate losses for BrainLM-EEG-Microstate. Panels within each row share a vertical scale.

size of 512 used here, this resulted in a learning rate of 3×10^{-4} . A weight decay of 0.05 was applied to weight parameters, while bias and normalisation parameters were excluded from weight decay. The learning rate increased linearly from zero during the first 40 epochs and subsequently decreased towards zero using cosine decay. Gradients were clipped to a maximum global norm of 1.0.

Models were pretrained for a maximum of 500 epochs. Early stopping was applied when the validation reconstruction loss did not improve for 30 consecutive epochs, and the checkpoint with the lowest validation reconstruction loss was retained for downstream evaluation. Using the same reconstruction-based stopping and checkpoint-selection criterion for all variants kept the model selection procedure consistent across the comparison.

A diagnostic linear probe was performed every 50 epochs to monitor the development

of the learned representations. At each interval, the encoder was frozen and a linear classification head was trained on the Alzheimer’s disease versus healthy control task using the same participant level partitions as the corresponding downstream evaluation. The probe was used only as a training diagnostic; its performance did not affect encoder optimisation, early stopping, or checkpoint selection and was therefore not included in the reported downstream results. The pretraining loss trajectories are shown in Figure 4.9.

4.6 Downstream Evaluation

The downstream evaluation assessed whether the representations learned during self-supervised pretraining transferred to diagnostic classification tasks. Four binary classification tasks were considered: Alzheimer’s disease (AD) versus healthy controls, frontotemporal dementia (FTD) versus healthy controls, first-episode psychosis (FEP) versus healthy controls, and major depressive disorder (MDD) versus healthy controls.

4.6.1 Subject Level Prediction and Evaluation Metrics

All downstream performance was evaluated at the participant level because multiple EEG windows from the same participant are not independent observations. Let $p_{s,i}$ denote the predicted probability of the patient class for window i from participant s , and let N_s denote the number of windows from that participant. The continuous participant level score was calculated as the mean patient class probability across windows,

$$\bar{p}_s = \frac{1}{N_s} \sum_{i=1}^{N_s} p_{s,i}. \quad (4.16)$$

For metrics requiring a hard class prediction, each window was assigned to the class with the higher predicted probability, and the participant level prediction was obtained by majority vote across its windows. In the case of an equal vote, the tie was resolved using the mean patient class probability $\bar{p}_s$.

Balanced accuracy (BAcc) and area under the precision-recall curve (AUPRC) were used as the primary evaluation metrics. Balanced accuracy was used because the downstream datasets were not perfectly class balanced, with the largest imbalance occurring in MDD. It gives equal weight to sensitivity and specificity and was calculated as

$$\text{BAcc} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right). \quad (4.17)$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively, with the patient group treated as the positive class.

AUPRC was computed using the average-precision formulation, with the patient group treated as the positive class. The participant level probability $\bar{p}_s$ was used as the continuous score for this calculation. Balanced accuracy therefore used the majority vote participant label, whereas AUPRC used the mean patient class probability across windows.

AUPRC was preferred to the area under the receiver operating characteristic curve (AUROC) because the clinical objective was not only to separate patients from healthy controls, but also to ensure that participants identified as having a disorder were likely to have it. AUPRC is more informative under class imbalance and is the more clinically relevant summary for this question. The average precision of the patient class alone was reported, rather than the mean of the patient and healthy class values, because averaging the two would have returned the chance level to 0.5 for every task and hidden exactly the sensitivity to class imbalance that motivated the choice of AUPRC.

4.6.2 Linear Probing and Finetuning

Two adaptation settings were evaluated. For linear probing, the pretrained encoder was frozen and a three-layer classification head was trained using the CLS token representation. This setting evaluated how informative the pretrained representation was for each downstream task without modifying the encoder.

For finetuning, both the pretrained encoder and the classification head were updated using the downstream training data. A learning rate of 5×10^{-5} was used to allow task specific adaptation while limiting the magnitude of updates to the pretrained encoder. Comparing linear probing and finetuning provided complementary measures of how well the pretrained representations transferred to the downstream tasks.

The same classification head was used in both settings, so they differ only in whether the encoder was updated. The architecture of the head and the complete optimisation settings for both are given in Appendix A.1.

4.6.3 Within-Dataset Cross-Validation

Because the downstream datasets contained relatively few participants, within-dataset evaluation was performed using five-fold cross-validation. The folds were constructed at the participant level using a fixed random seed and were stratified by diagnostic class. All windows from a participant were assigned to the same fold to prevent participant level leakage between training, validation, and test sets.

For each run, three folds were used for training, one for validation, and one for testing. The fold assignments were rotated across five runs so that each fold served as the test set once. This produced five trained models for each task and adaptation setting. Performance was evaluated on the held-out test participants and averaged across the five runs. This evaluation measured generalisation to unseen participants from the same dataset. The fold structure is illustrated in Figure 4.10.

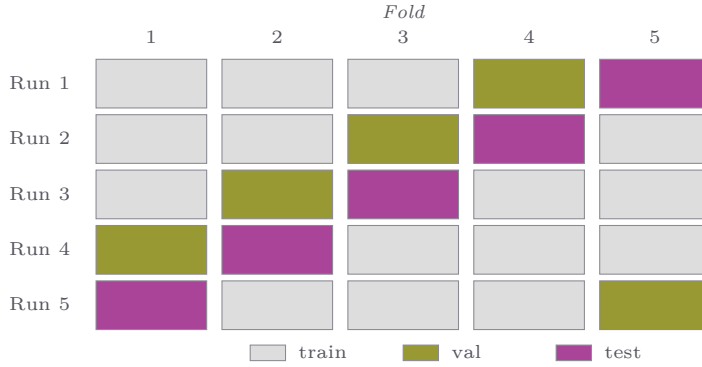


Figure 4.10: Five-fold participant level cross-validation. Fold assignment for each of the five cross-validation runs.

4.6.4 Leave-One-Subject-Out Evaluation

The five-fold experiments showed relatively large variation across folds for some tasks. A leave-one-subject-out (LOSO) evaluation was therefore additionally performed to reduce dependence on a particular five-fold partition and to obtain an out-of-sample prediction for each participant.

For a dataset containing N participants, each LOSO iteration used one participant as the test subject, one different participant as the validation subject, and the remaining $N - 2$ participants for training. Training continued according to the early-stopping criterion, and the checkpoint with the lowest validation loss was selected. The selected checkpoint was then evaluated on the held-out test participant. This procedure was

repeated until every participant had served as the test subject once. The resulting subject level predictions were pooled to calculate the final performance metrics. The procedure is illustrated in Figure 4.11.

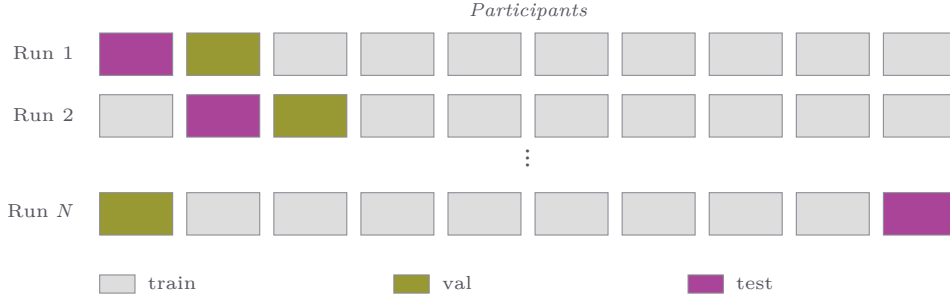


Figure 4.11: Leave-one-subject-out cross-validation. Participant assignment for three representative runs; the ellipsis marks the remaining runs up to run N .

4.6.5 Cross-Dataset Evaluation

Cross-dataset evaluation was used to examine transfer to data collected in a different dataset. The five models trained on the FEP dataset during cross-validation were each evaluated on the independent SCZ dataset, and their performance was averaged. In addition, a model trained using the full FEP development data was evaluated on SCZ. This provided an external evaluation of how well representations learned from FEP data transferred to a separate schizophrenia dataset. This transfer setting is illustrated in Figure 4.12.

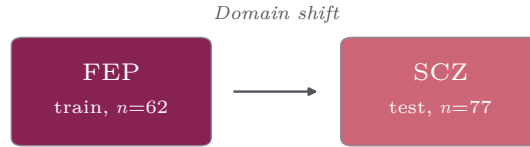


Figure 4.12: Cross-dataset transfer. Models trained on FEP ($n = 62$) are evaluated on the independently acquired SCZ cohort ($n = 77$).

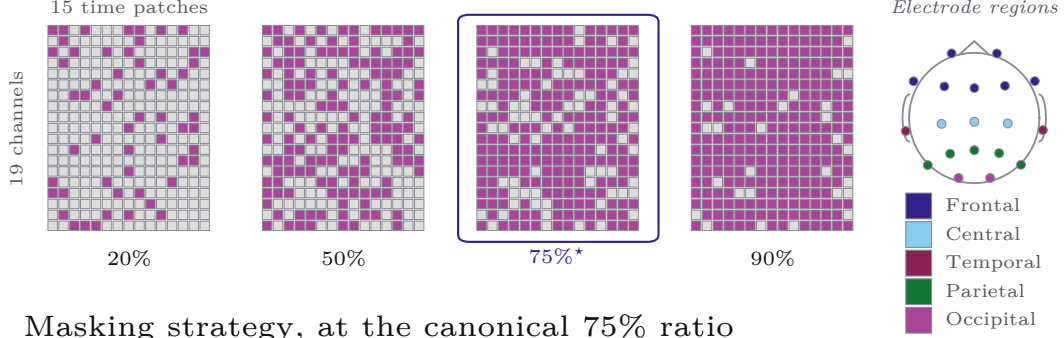
4.7 Ablation Studies and Model Analysis

In addition to the downstream evaluation, a set of ablation and analysis experiments was conducted to examine how selected modelling choices affected model behaviour and downstream performance. These experiments considered the masking configuration used during pretraining, the information available at different encoder layers, the contribution

of the CLS token to reconstruction, and the extent to which reconstructed EEG retained information relevant to downstream classification.

4.7.1 Masking Strategy and Masking Ratio

(a) Masking ratio, random channel $\times$ time masking



(b) Masking strategy, at the canonical 75% ratio

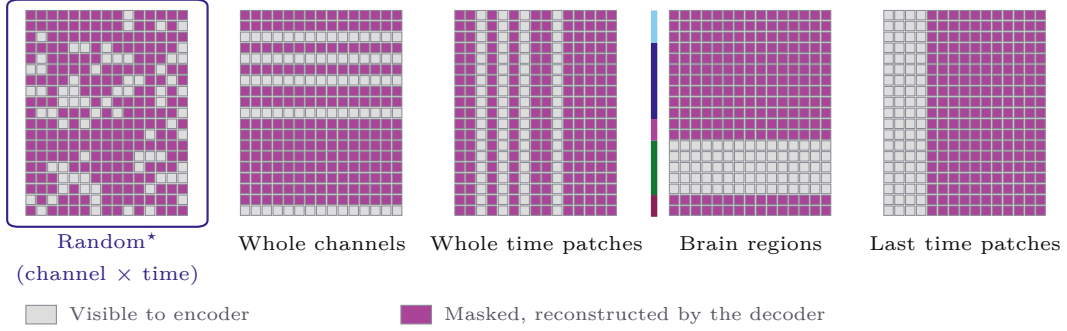


Figure 4.13: The nine pretraining masking variants, drawn on the 19×15 grid of spatiotemporal tokens produced by a single 15-second window: rows are the 19 channels in the order the model receives them, columns are the 15 one-second temporal patches, and shaded cells are the tokens withheld from the encoder and reconstructed by the decoder. Starred panels mark the canonical setting used for BrainLM-EEG elsewhere in this work. Each pattern is one representative random draw under a fixed seed. In the brain region panel, whole regions are masked at random until the closest total under the nominal ratio is reached; the colour bar at that panel’s left edge marks the region blocks and is keyed to the scalp map in (a).

The baseline BrainLM-EEG configuration described in Section 4.3.1 used a masking ratio of 75% with random spatiotemporal masking. To examine the effect of the amount of masked input, additional models were pretrained using masking ratios of 20%, 50%, and 90%, while the remaining pretraining settings were kept unchanged.

The masking structure was also varied to examine whether different forms of missing

information affected representation learning. The evaluated strategies included masking complete channels across time, masking randomly selected temporal regions across all channels, masking the final portion of the input window across all channels, and region-wise masking in which electrodes were grouped by anatomical brain region and all electrodes within a selected region were masked together. These strategies were intended to represent different forms of missing spatial or temporal information, including missing electrodes and forecasting-like reconstruction. All variants were evaluated using the downstream protocol described in Section 4.6. The nine resulting variants are shown in Figure 4.13.

4.7.2 Layer-Wise Representation Analysis

All encoder stages frozen

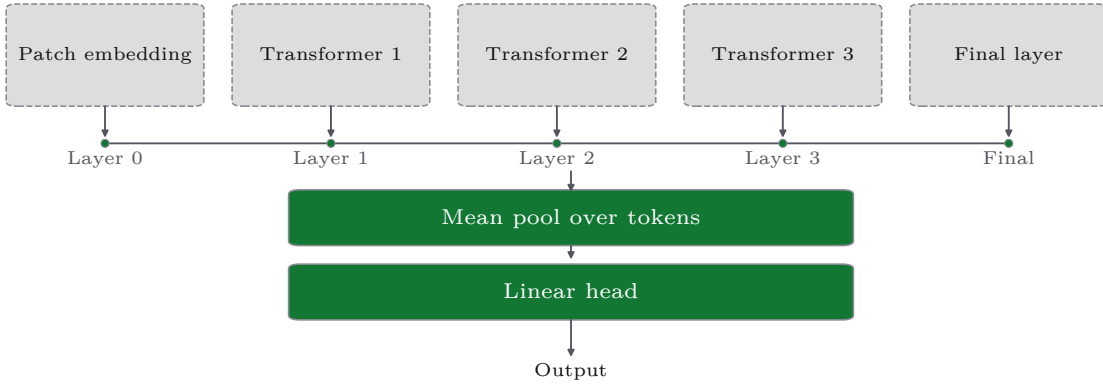


Figure 4.14: Layer-wise probing of the pretrained BrainLM-EEG encoder with frozen encoder weights. A separate linear head is trained on token-averaged representations from each encoder stage; layer 0 denotes the patch embedding.

A layer-wise probing experiment was performed to examine how task relevant information was distributed across the encoder. After pretraining, the encoder was frozen and representations were extracted from each encoder stage. For each stage, the token representations were averaged across the sequence to obtain a single representation for each EEG window. A separate linear classification head was then trained on the resulting representation.

Each encoder stage was evaluated separately using the same downstream evaluation protocol described in Section 4.6. This analysis was used to compare the amount of downstream discriminative information available at different depths of the encoder and to determine how this information emerges and evolves throughout the learned representation hierarchy. The overall probing procedure is illustrated in Figure 4.14.

4.7.3 CLS Token Contribution to Reconstruction

BrainLM-EEG uses a masked autoencoder architecture in which the encoder maps an EEG window to latent token representations, including a CLS token, and the decoder reconstructs the masked input from these representations. Although the CLS token is used for downstream classification, it is not directly supervised by a classification objective during self-supervised pretraining. An additional analysis was therefore performed to examine whether the decoder made use of the CLS representation during reconstruction.

EEG windows from the pretraining validation set were first passed through the pretrained encoder. Reconstruction was then performed under three conditions: (1) using the encoder output without modification, (2) replacing the CLS representation with zeros, and (3) replacing the CLS representation with Gaussian noise. All other latent token representations were left unchanged. Reconstruction error was measured using the same mean squared error objective used during pretraining.

Because the same EEG windows were evaluated under each condition, paired Wilcoxon signed-rank tests were used to compare the reconstruction losses obtained with the unmodified CLS token and the perturbed CLS conditions. An increase in reconstruction error after perturbing the CLS token would indicate that information carried by this token contributes to the decoder’s reconstruction. This procedure is summarised in Figure 4.15.

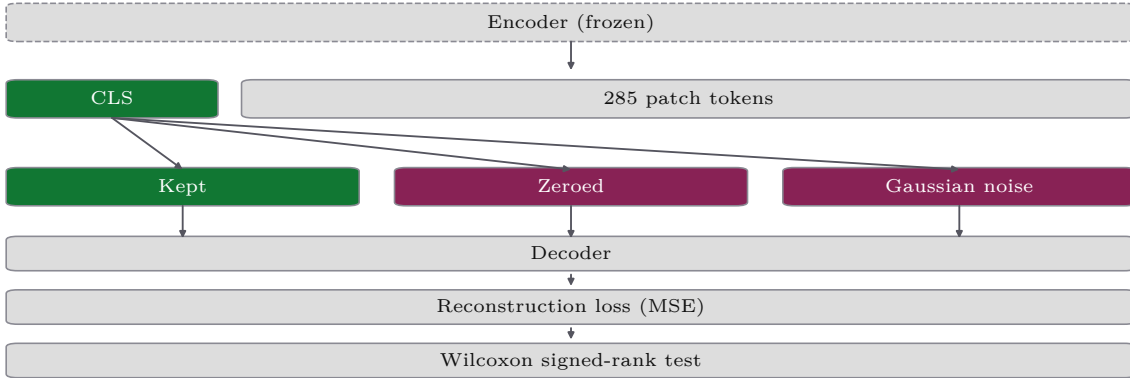


Figure 4.15: CLS token test on the pretrained BrainLM-EEG encoder with frozen encoder weights. Each window is encoded once, and reconstruction is evaluated with the CLS latent kept, zeroed, or replaced by Gaussian noise. Paired losses are compared using a Wilcoxon signed-rank test; increased loss under corruption indicates CLS-token use.

4.7.4 Analysis of Reconstructed EEG

A final experiment examined whether EEG reconstructed by the pretrained encoder-decoder retained information relevant to downstream classification. EEG windows from the downstream datasets were passed through the pretrained model to obtain reconstructed signals. The same baseline classifiers used elsewhere in the study, including LDA and ATCNet, were then trained separately using either the original EEG windows or the corresponding reconstructed windows.

The models were evaluated using the same participant level downstream protocol described in Section 4.6, and performance obtained from original and reconstructed EEG was compared. This analysis assessed the extent to which the reconstructed signals retained information used by the downstream classifiers to distinguish patients from healthy controls. Figure 4.16 summarises this comparison.

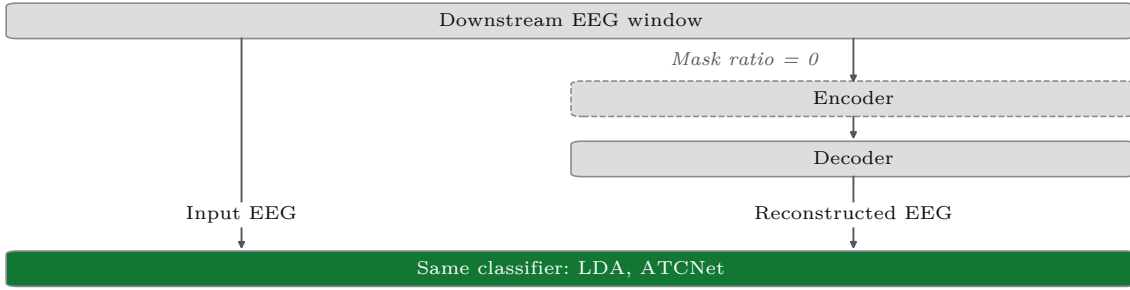


Figure 4.16: Reconstruction as features using the pretrained BrainLM-EEG encoder-decoder with frozen weights. The same classifier is trained on the original and reconstructed windows (masking ratio = 0) to assess whether class discriminative information is preserved.

5 Results

This chapter evaluates not only whether the learned EEG representations support clinical classification, but also the conditions under which that information remains useful. The analysis therefore proceeds from within-dataset five-fold cross-validation to leave-one-subject-out (LOSO) evaluation and then to cross-dataset transfer from FEP to SCZ. This progression separates performance on unseen participants from robustness to a change of dataset. The remaining experiments examine how the representations are used and where task relevant information is expressed, through finetuning, masking ablations, layer-wise analysis, the CLS perturbation experiment, and classification using reconstructed EEG. Unless otherwise stated, all downstream metrics are reported at the participant level using the procedures defined in Section 4.6.

5.1 How to Read the Results

Three points are important when interpreting the results in this chapter.

First, balanced accuracy (BAcc) and area under the precision-recall curve (AUPRC) were both calculated at the participant level, but by different forms of aggregation: BAcc from the majority vote label and AUPRC from the mean patient class probability (Section 4.6.1). They therefore need not change in the same direction. A model may rank participants well by their mean probabilities while still assigning some of them to the wrong class at the decision threshold.

Second, AUPRC was computed using average precision with the patient group as the positive class (Section 4.6.1). Its prevalence reference therefore differs across tasks. Table 5.1 lists the downstream cohort sizes currently reported in the study and the corresponding reference levels. For example, the reference AUPRC is 26.5% for MDD and 61.3% for FEP. BAcc has a nominal chance level of 50% for each binary task. Accordingly, AUPRC comparisons in this chapter are made primarily between models evaluated on the same task. BAcc has a common nominal chance level, but comparisons across tasks should still be interpreted cautiously because the datasets and classification problems differ.

Table 5.1: Downstream cohort sizes and chance levels of the two reported metrics.

Task	Subjects	Patients / HC	AUPRC chance	BAcc chance
AD versus HC	52	29 / 23	55.8	50.0
FTD versus HC	41	18 / 23	43.9	50.0
FEP versus HC	62	38 / 24	61.3	50.0
MDD versus HC	98	26 / 72	26.5	50.0
SCZ (external test)	77	30 / 47	39.0	50.0

Subject counts are the downstream evaluation pools remaining after the pretraining reserve participants were removed (Section 4.1). The SCZ cohort was never used for training and appears only as the external test set in Section 5.4.

Third, no inferential significance tests were performed for comparisons among the downstream models, so these comparisons are descriptive. For the five-fold results, the reported standard deviation describes variation across the five folds rather than statistical uncertainty about a population effect. For LOSO, predictions were pooled across test participants to produce one BAcc and one AUPRC value, so no fold level standard deviation is available (Section 4.6.4). Bold and underlined values in the result tables mark only the highest and second-highest numerical values and should not be interpreted as evidence of statistically significant differences.

5.2 Within-Dataset Performance

Table 5.2: In-domain classification performance on four clinical EEG tasks (subject level, mean $\pm$ std over five-fold cross-validation).

	AD		FTD		FEP		MDD	
	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$
LDA	95.0 $\pm$ 5.4	87.3 $\pm$ 10.6	82.0 $\pm$ 12.6	77.5 $\pm$ 13.3	72.3 $\pm$ 2.5	57.9 $\pm$ 7.2	35.3 $\pm$ 11.8	49.4 $\pm$ 11.5
SVM	93.0 $\pm$ 7.0	80.8 $\pm$ 11.0	82.1 $\pm$ 13.3	66.3 $\pm$ 23.6	75.4 $\pm$ 14.3	56.8 $\pm$ 19.4	38.7 $\pm$ 13.2	<u>55.0 $\pm$ 10.8</u>
XGBoost	95.4 $\pm$ 5.3	85.3 $\pm$ 11.1	79.3 $\pm$ 16.0	68.3 $\pm$ 14.1	70.4 $\pm$ 11.1	51.1 $\pm$ 10.3	33.8 $\pm$ 11.6	54.9 $\pm$ 6.8
EEGNeX	95.3 $\pm$ 4.8	80.8 $\pm$ 17.4	89.8 $\pm$ 15.6	81.3 $\pm$ 15.7	67.9 $\pm$ 12.3	52.5 $\pm$ 13.6	40.8 $\pm$ 14.1	50.0 $\pm$ 0.0
EEGConformer	94.3 $\pm$ 3.3	67.8 $\pm$ 10.9	83.5 $\pm$ 7.7	64.2 $\pm$ 13.8	73.1 $\pm$ 6.9	53.2 $\pm$ 9.0	<u>45.7 $\pm$ 22.1</u>	50.3 $\pm$ 0.6
ATCNet	93.7 $\pm$ 6.8	85.3 $\pm$ 13.7	86.2 $\pm$ 14.2	79.2 $\pm$ 10.5	77.1 $\pm$ 7.3	60.4 $\pm$ 7.8	46.0 $\pm$ 17.5	48.0 $\pm$ 4.0
LaBraM (LP)	89.9 $\pm$ 11.2	73.5 $\pm$ 18.2	69.5 $\pm$ 21.5	73.5 $\pm$ 19.0	68.2 $\pm$ 5.5	51.5 $\pm$ 8.9	36.4 $\pm$ 6.9	50.0 $\pm$ 0.0
CBraMod (LP)	92.1 $\pm$ 6.6	80.0 $\pm$ 7.8	75.8 $\pm$ 18.7	68.0 $\pm$ 14.8	68.1 $\pm$ 3.5	47.1 $\pm$ 7.1	28.5 $\pm$ 7.4	49.3 $\pm$ 1.3
REVE (LP)	<u>97.1 $\pm$ 2.5</u>	81.7 $\pm$ 12.0	91.3 $\pm$ 9.0	<u>80.5 $\pm$ 12.0</u>	81.0 $\pm$ 4.8	<u>58.5 $\pm$ 3.2</u>	33.3 $\pm$ 10.2	52.9 $\pm$ 3.9
BrainLM-EEG (LP)	95.9 $\pm$ 5.7	78.8 $\pm$ 15.6	83.8 $\pm$ 16.3	69.2 $\pm$ 15.9	77.0 $\pm$ 6.0	56.5 $\pm$ 9.7	30.4 $\pm$ 7.9	52.6 $\pm$ 5.1
BrainLM-EEG-RoPE (LP)	92.4 $\pm$ 6.7	81.2 $\pm$ 13.3	73.0 $\pm$ 16.3	48.5 $\pm$ 15.2	74.6 $\pm$ 4.5	52.8 $\pm$ 12.0	36.3 $\pm$ 12.8	50.0 $\pm$ 0.0
BrainLM-EEG-Microstate (LP)	100.0 $\pm$ 0.0	87.3 $\pm$ 8.9	87.0 $\pm$ 14.5	76.3 $\pm$ 12.7	<u>80.5 $\pm$ 5.1</u>	54.5 $\pm$ 6.3	32.5 $\pm$ 12.2	55.3 $\pm$ 6.8

AD: Alzheimer’s disease; FTD: frontotemporal dementia; FEP: first-episode psychosis; MDD: major depressive disorder; all versus healthy controls. LP: linear probe. Values are subject level $\times 100$. **Bold**: best per column; underline: second best.

Table 5.2 reports performance under the five-fold within-dataset protocol described in Section 4.6.3. The first pattern is task dependence: performance differed substantially across the four clinical tasks, and the model ordering also changed from one task to another. The results are therefore interpreted first within each task rather than as a single overall ranking of model families.

For AD versus HC, all model families achieved an AUPRC of at least 89.9%, compared with an AUPRC reference of 55.8%. BAcc ranged from 67.8% for EEGConformer to 87.3% for BrainLM-EEG-Microstate and LDA, which both reported the highest BAcc, while the Microstate variant also achieved 100% AUPRC. These values are not contradictory because AUPRC was calculated from mean participant level probabilities, whereas BAcc used majority vote class predictions (Section 5.1). This AUPRC therefore indicates perfect ranking within the evaluated folds, not perfect classification at the decision threshold. Because both the proposed models and the comparison baselines performed strongly, these results do not provide evidence of a specific advantage for self-supervised pretraining on this task (Figure 5.1).

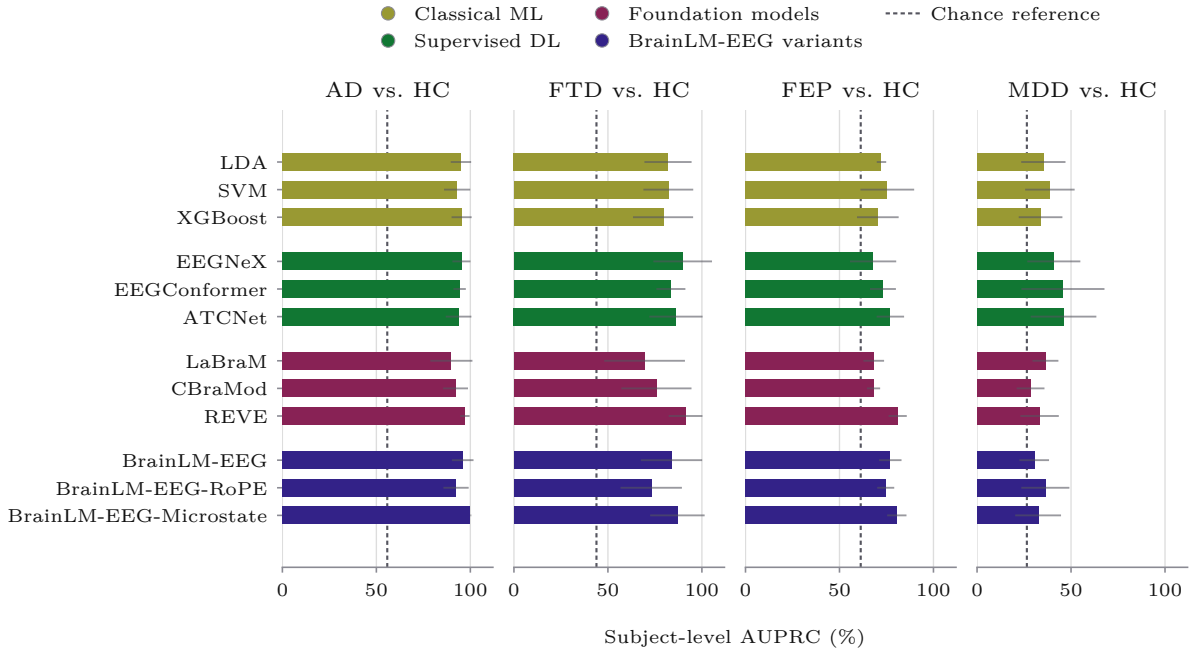


Figure 5.1: Subject level AUPRC for each task. Bars are grouped by model family; error bars denote fold standard deviation, and dashed lines indicate the task specific AUPRC chance reference.

For FTD versus HC, EEGNeX and REVE achieved the highest BAcc values, at 81.3% and 80.5%, respectively. BrainLM-EEG-RoPE achieved 48.5% BAcc, below the nominal 50% chance level. The model ordering changed under LOSO, as reported in Section 5.3. For FEP versus HC, REVE reported the highest AUPRC at 81.0%, followed by BrainLM-EEG-Microstate at 80.5%. ATCNet reported the highest BAcc at 60.4%, and CBraMod

reported the lowest at 47.1%. The highest FEP AUPRC was therefore 19.7 percentage points above the 61.3% prevalence reference for this task.

For MDD versus HC, ATCNet and EEGConformer achieved AUPRC values of 46.0% and 45.7%, respectively, above the 26.5% prevalence reference. In contrast, BAcc ranged from 48.0% to 55.3% across models. Several models had BAcc values very close to 50%, including values of exactly 50.0% for EEGNeX, LaBraM, and BrainLM-EEG-RoPE. Overall, some models showed above-reference ranking on MDD, whereas thresholded participant level classification remained close to chance. These results do not determine whether this pattern reflects the dataset, the learned representations, the classification heads, or the chosen decision rule.

The within-dataset results establish a task-dependent baseline. AD was strongly decodable across model families, whereas the relative model ordering was less stable on FTD and FEP, and MDD showed a clearer separation between ranking performance and thresholded classification. Within the BrainLM-EEG family, the Microstate variant achieved higher numerical performance than the base model in several task metric comparisons, whereas the RoPE variant showed a less consistent pattern. However, neither modification was consistently better across tasks or metrics, and the observed fold-to-fold variation further limits the interpretation of small numerical differences. The next analysis therefore examines whether these patterns remain when each participant receives an out-of-sample prediction under LOSO.

5.3 Leave-One-Subject-Out Evaluation

The LOSO procedure is described in Section 4.6.4. Each iteration used one participant for testing, one for validation, and the remaining participants for training. After every participant had served as the test participant once, the out-of-sample predictions were pooled to calculate a single BAcc and AUPRC value for each model and task. Table 5.3 reports these results. Because each LOSO test fold contains only one participant, BAcc and AUPRC cannot be meaningfully calculated at the fold level, in particular balanced accuracy requires observations from both classes. The metrics were therefore calculated once from the pooled out-of-sample predictions.

For AD versus HC, all model families again achieved relatively high performance. REVE had the highest values, with 98.0% AUPRC and 90.9% BAcc, while several other models, including the BrainLM-EEG variants, were within a few percentage points. For FTD versus HC, the strongest numerical results depended on the metric. REVE achieved the highest AUPRC at 95.5%, followed by EEGConformer at 95.0% and BrainLM-EEG-Microstate at 94.6%. For BAcc, BrainLM-EEG-RoPE and BrainLM-EEG-Microstate achieved 92.9% and 92.3%, respectively, compared with 91.7% for REVE.

Table 5.3: Leave-one-subject-out (LOSO) evaluation on four clinical EEG tasks.

	AD		FTD		FEP		MDD	
	AUPRC $\uparrow$	BACC $\uparrow$	AUPRC $\uparrow$	BACC $\uparrow$	AUPRC $\uparrow$	BACC $\uparrow$	AUPRC $\uparrow$	BACC $\uparrow$
LDA	92.2	85.8	85.0	78.4	78.9	<u>66.0</u>	34.2	57.2
SVM	93.4	85.3	75.8	71.3	65.6	55.8	30.7	50.5
XGBoost	94.9	82.7	77.1	62.9	67.0	59.5	38.6	55.1
EEGNeX	92.9	83.1	86.9	80.2	63.1	50.5	<u>41.9</u>	54.1
EEGConformer	96.8	86.6	<u>95.0</u>	77.8	60.7	44.1	34.7	<u>58.8</u>
ATCNet	92.8	<u>87.0</u>	89.2	79.0	78.9	68.4	37.4	54.6
LaBraM (LP)	93.1	85.3	88.7	79.6	54.4	43.9	42.5	53.2
CBraMod (LP)	95.9	85.7	80.9	75.2	56.5	43.3	34.4	50.5
REVE (LP)	98.0	90.9	95.5	91.7	62.2	45.4	37.7	61.4
BrainLM-EEG (LP)	94.7	81.4	86.2	81.8	68.3	61.1	32.9	49.5
BrainLM-EEG-RoPE (LP)	96.1	<u>87.0</u>	92.3	92.9	<u>68.7</u>	43.9	36.3	51.1
BrainLM-EEG-Microstate (LP)	<u>96.9</u>	85.3	94.6	<u>92.3</u>	62.8	56.9	37.9	51.6

AD: Alzheimer’s disease; FTD: frontotemporal dementia; FEP: first-episode psychosis; MDD: major depressive disorder; all versus healthy controls. LP: linear probe. Values are subject level $\times 100$. **Bold**: best per column; underline: second best.

For FEP versus HC, several pretrained models achieved lower AUPRC values under LOSO than under five-fold evaluation. ATCNet and LDA each achieved 78.9% AUPRC, while a number of self-supervised and pretrained foundation models reported lower values. For MDD, AUPRC ranged from 30.7% to 42.5%, with LaBraM highest at 42.5% and EEGNeX second at 41.9%, whereas BACC ranged from 49.5% to 61.4%, with multiple models remaining close to the nominal 50% chance level.

Within the proposed model family, both RoPE and Microstate achieved higher numerical AUPRC and BACC than the base BrainLM-EEG model on AD and FTD. The same pattern was not observed consistently for FEP or MDD, indicating that the relative benefit of the proposed modifications was task dependent.

A particularly large protocol-dependent change occurred for FTD. BrainLM-EEG-RoPE changed from 73.0% AUPRC and 48.5% BACC under five-fold cross-validation to 92.3% AUPRC and 92.9% BACC under LOSO. BrainLM-EEG-Microstate changed from 87.0% AUPRC and 76.3% BACC to 94.6% AUPRC and 92.3% BACC. Because the FTD evaluation pool contained only 41 participants, the individual five-fold test sets were small. The marked difference between protocols therefore indicates that the reported performance and model ordering were sensitive to the resampling scheme in this dataset. Pooling the LOSO predictions removes fold-to-fold reporting variation, but it does not remove uncertainty associated with the small cohort.

Changes between protocols did not consistently favour the proposed models. On FEP, REVE changed from 81.0% AUPRC under five-fold evaluation to 62.2% under LOSO, while LaBraM changed from 68.2% to 54.4%. Under LOSO, LDA achieved a higher FEP

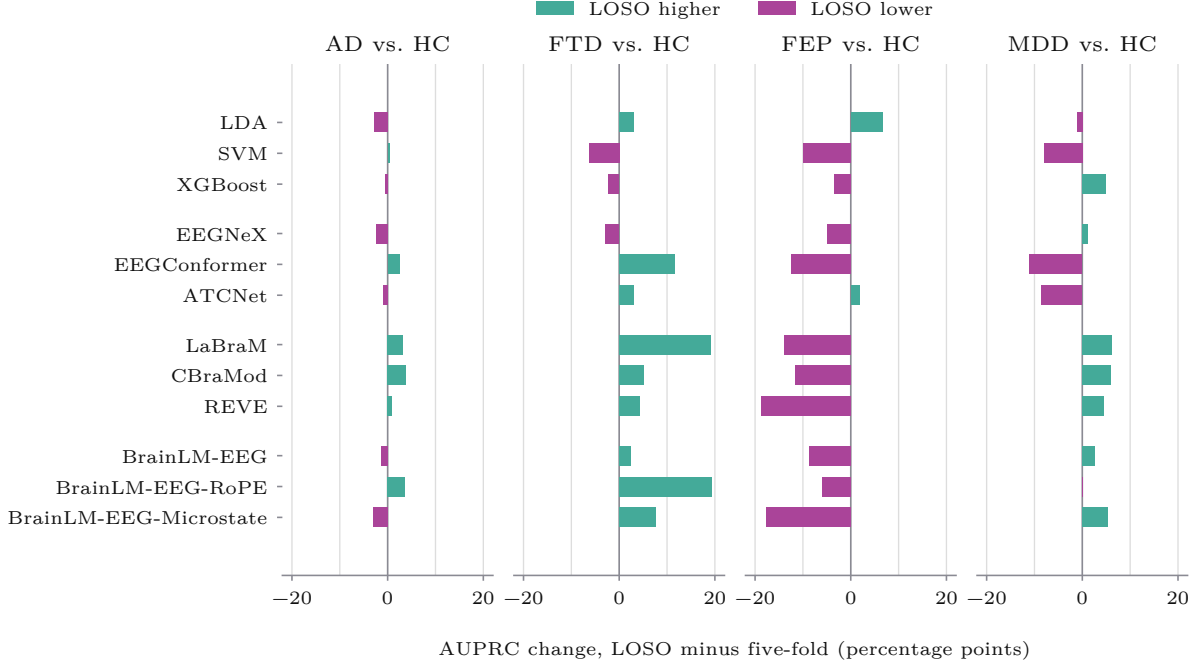


Figure 5.2: Difference in subject level AUPRC between pooled LOSO and five-fold cross-validation for each model and task. Bars show LOSO minus five-fold AUPRC.

AUPRC than the evaluated foundation models. For some model task combinations, the differences between evaluation protocols were therefore as large as or larger than the numerical differences between model families. Figure 5.2 shows the protocol difference for every model and task.

The LOSO analysis therefore reinforces the sensitivity of the reported model comparisons to the evaluation protocol. The BrainLM-EEG variants were among the stronger models on FTD when predictions from all held-out participants were pooled, demonstrating that the pretrained representations supported strong participant level decoding on this task. At these cohort sizes, which model appears to perform best can depend on which participants are included in the training and test sets.

5.4 Cross-Dataset Generalisation from FEP to SCZ

Cross-dataset evaluation examined whether models developed on FEP transferred to an independently collected SCZ cohort, following Section 4.6.5. The two datasets were collected independently and differed in cohort and acquisition characteristics (Salisbury et al., 2022; Racz et al., 2025). The comparison therefore reflects a combined dataset and acquisition shift rather than isolating any single source of distribution shift.

Table 5.4: Cross-dataset transfer from FEP to SCZ, alongside in-domain FEP performance (subject level).

	FEP		FEP → SCZ (five-fold)		FEP → SCZ (all-data)	
	AUPRC ↑	BACC ↑	AUPRC ↑	BACC ↑	AUPRC ↑	BACC ↑
LDA	72.3 ±2.5	57.9 ±7.2	40.2 ±4.0	50.0 ±3.6	39.2	55.2
SVM	75.4 ±14.3	56.8 ±19.4	43.2 ±4.8	53.8 ±2.7	54.4	55.3
XGBoost	70.4 ±11.1	51.1 ±10.3	42.4 ±3.0	47.4 ±2.2	41.8	51.4
EEGNeX	67.9 ±12.3	52.5 ±13.6	43.3 ±8.7	50.8 ±2.9	57.6	63.4
EEGConformer	73.1 ±6.9	53.2 ±9.0	47.2 ±2.3	51.7 ±2.2	48.4	55.7
ATCNet	77.1 ±7.3	60.4 ±7.8	54.4 ±6.9	<u>56.5 ±3.9</u>	49.5	<u>62.7</u>
LaBraM (LP)	68.2 ±5.5	51.5 ±8.9	48.5 ±4.8	51.3 ±3.4	<u>56.8</u>	57.6
CBraMod (LP)	68.1 ±3.5	47.1 ±7.1	43.0 ±6.7	50.2 ±3.3	42.9	51.5
REVE (LP)	81.0 ±4.8	<u>58.5 ±3.2</u>	46.1 ±5.5	55.8 ±3.0	45.2	57.6
BrainLM-EEG (LP)	77.0 ±6.0	56.5 ±9.7	50.5 ±5.8	52.3 ±2.4	42.2	56.1
BrainLM-EEG-RoPE (LP)	74.6 ±4.5	52.8 ±12.0	52.7 ±4.7	55.7 ±4.7	47.4	61.3
BrainLM-EEG-Microstate (LP)	<u>80.5 ±5.1</u>	54.5 ±6.3	<u>53.1 ±2.4</u>	58.1 ±4.5	46.2	49.5

FEP: first-episode psychosis; SCZ: schizophrenia; all versus healthy controls. LP: linear probe. Values are subject level $\times 100$. **Bold**: best per column; underline: second best.

Table 5.4 reports three settings. The first column gives within-dataset FEP performance from five-fold cross-validation. In the second setting, each of the five FEP-trained checkpoints was evaluated directly on SCZ and the resulting metrics were averaged. In the third setting, a model trained on the complete FEP data was evaluated once on SCZ.

AUPRC decreased for every model when the FEP-trained models were evaluated on SCZ under either transfer setting. In the five-checkpoint setting, ATCNet achieved the highest AUPRC at 54.4%, followed by BrainLM-EEG-Microstate at 53.1% and BrainLM-EEG-RoPE at 52.7%, while LDA achieved the lowest AUPRC of 40.2%. The externally pretrained models showed a similar reduction, with REVE, LaBraM, and CBraMod achieving 46.1%, 48.5%, and 43.0% AUPRC, respectively, in this setting. In the full-FEP training setting, EEGNeX achieved the highest AUPRC and BACC, at 57.6% and 63.4%, respectively. Across the transfer results, BACC ranged from 47.4% to 63.4%. Using the SCZ class counts currently reported in Table 5.1, the AUPRC prevalence reference is 39.0%. Figure 5.3 places these transfer values beside each model’s in-domain FEP result.

Within the BrainLM-EEG family, both variants achieved higher AUPRC than the base model in both transfer settings, whereas the corresponding BACC differences were not consistently positive. No model family showed a consistent numerical robustness advantage across the two transfer settings. For example, EEGNeX changed from 43.3% AUPRC in the five-checkpoint setting to 57.6% when trained on the full-FEP data, whereas BrainLM-EEG-Microstate changed from 53.1% to 46.2%. Because the full-data

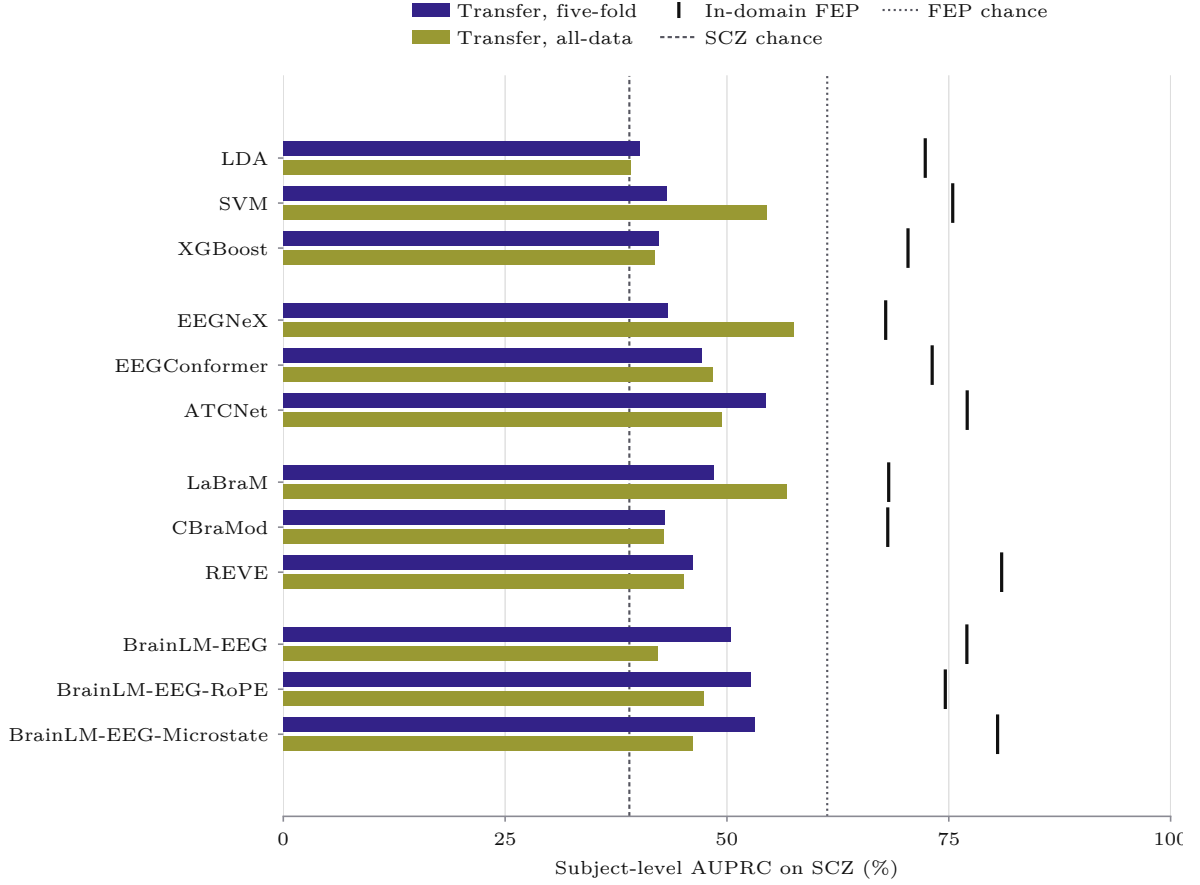


Figure 5.3: Subject level AUPRC on SCZ for five-fold and all-data FEP transfer checkpoints. Vertical rules indicate each model’s in-domain FEP AUPRC; dashed and dotted lines denote the SCZ and FEP chance references, respectively.

setting is based on a single trained checkpoint, it provides no estimate of run-to-run variation. Differences between the two transfer settings should therefore be interpreted descriptively.

The FEP to SCZ experiment therefore establishes an important boundary on the preceding within-dataset results. Models that produced out-of-sample predictions for unseen participants from the source dataset did not retain the same ranking performance after transfer to the external dataset. The transfer was limited and variable across model families, and no family showed a consistent robustness advantage across the two transfer settings. This does not establish that the learned representations cannot transfer across datasets more generally. Because FEP performance was itself protocol dependent, the experiment also cannot separate the effect of dataset shift from limitations of the source task model. A stronger assessment would require additional external dataset pairs.

5.5 Linear Probing versus Finetuning

Table 5.5: Linear probing versus full finetuning of BrainLM-EEG model variants under LOSO evaluation.

	AD		FTD		FEP		MDD	
	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$
BrainLM-EEG (LP)	94.7	81.4	86.2	81.8	<u>68.3</u>	61.1	32.9	49.5
BrainLM-EEG-RoPE (LP)	<u>96.1</u>	87.0	<u>92.3</u>	92.9	68.7	43.9	<u>36.3</u>	51.1
BrainLM-EEG-Microstate (LP)	96.9	<u>85.3</u>	94.6	<u>92.3</u>	62.8	<u>56.9</u>	37.9	<u>51.6</u>
BrainLM-EEG (FT)	93.3	84.9	87.4	73.4	62.8	56.7	31.7	51.1
BrainLM-EEG-RoPE (FT)	88.0	80.5	82.6	69.1	61.4	51.4	29.6	51.8
BrainLM-EEG-Microstate (FT)	93.7	84.0	86.7	76.2	60.6	56.1	26.7	49.1

AD: Alzheimer’s disease; FTD: frontotemporal dementia; FEP: first-episode psychosis; MDD: major depressive disorder; all versus healthy controls. LP: linear probe; FT: full finetuning. Values are subject level $\times 100$. **Bold**: best per column; underline: second best.

The preceding three sections established how far the learned representations generalise. The remaining sections examine how they are used and where the task relevant information within them is expressed, beginning with the choice of adaptation strategy. Linear probing evaluates the frozen pretrained representation, whereas finetuning additionally updates the encoder using downstream labels (Section 4.6.2). Table 5.5 compares the two adaptation settings for the BrainLM-EEG variants under LOSO.

Finetuning produced higher values in only 5 of the 24 metric comparisons in the table. For AUPRC, it improved 1 of 12 comparisons: the base BrainLM-EEG model on FTD increased from 86.2% to 87.4%. Some BAcc values also increased, including those for BrainLM-EEG on AD and BrainLM-EEG-RoPE on FEP and MDD. However, several of the largest absolute changes were decreases. On FTD, BAcc for BrainLM-EEG-RoPE decreased from 92.9% to 69.1%, and BAcc for BrainLM-EEG-Microstate decreased from 92.3% to 76.2%. On MDD, AUPRC for BrainLM-EEG-Microstate decreased from 37.9% to 26.7%. These per-task changes are shown in Figure 5.4.

The downstream evaluation pools were small, containing 52, 41, 62, and 98 participants for AD, FTD, FEP, and MDD, respectively. Compared with linear probing, finetuning therefore updated substantially more parameters using the same limited labelled cohorts. This is consistent with the possibility of overfitting or unstable adaptation, but the experiment does not isolate the cause of the observed degradation. Only the finetuning configuration described in Section 4.6.2 was evaluated; no per-task learning rate search, partial unfreezing, or layer-wise learning rate schedule was compared.

Within the evaluated configuration, the more reliable use of the pretrained encoder was therefore as a frozen representation rather than through full-model adaptation: finetuning did not consistently improve over linear probing under the hyperparameters and data splits used in this study. This result does not show that finetuning is generally unsuitable

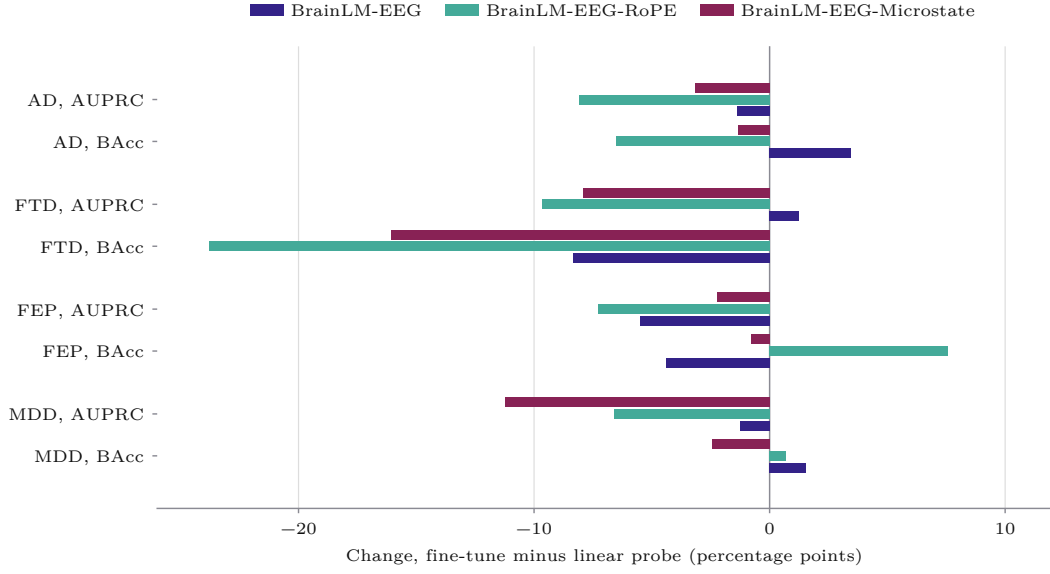


Figure 5.4: Change in AUPRC and balanced accuracy after finetuning, computed as finetuned minus linear-probed performance. Bars show the change for each proposed model, with negative values indicating better performance under linear probing.

for these tasks, but it identifies linear probing as the stronger adaptation strategy in the experiments reported here.

5.6 Masking Ratio and Masking Strategy

The masking ablation examined whether downstream performance changed with the amount and structure of masking used during BrainLM-EEG pretraining (Section 4.7.1). The analysis used the base BrainLM-EEG model with linear probing and five-fold within-dataset evaluation. Table 5.6 reports both comparisons. MDD was not included in this ablation because its downstream BAcc was close to chance in the preceding evaluations.

For the masking ratio comparison, random masking was retained while the ratio was varied. The 75% and 90% configurations produced higher values in five of the six reported task metric columns. The exception was FTD AUPRC when comparing 20% with the canonical 75% setting: 20% masking achieved 88.5%, whereas 75% masking achieved 82.8%. The highest FTD AUPRC in the masking ratio comparison was 91.0% at 90% masking. These results indicate that high masking ratios did not prevent the model from learning representations that supported the evaluated downstream tasks. They do not, however, establish that high masking ratios are generally optimal for EEG.

Table 5.6: Ablation of the BrainLM-EEG pretraining objective: masking ratio (upper block, random masking) and masking strategy (lower block, 75% masking ratio), under linear probing.

	AD		FTD		FEP	
	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$
Random, 20%	86.2 $\pm$ 16.1	66.0 $\pm$ 9.6	88.5 $\pm$ 7.1	69.8 $\pm$ 10.9	62.1 $\pm$ 13.1	45.9 $\pm$ 5.6
Random, 50%	89.7 $\pm$ 11.5	74.0 $\pm$ 15.9	81.1 $\pm$ 6.5	60.5 $\pm$ 12.9	67.6 $\pm$ 10.7	45.8 $\pm$ 8.3
Random, 75%*	<u>92.6 $\pm$ 8.0</u>	85.2 $\pm$ 4.0	82.8 $\pm$ 6.5	<u>74.7 $\pm$ 14.3</u>	73.6 $\pm$ 15.4	<u>54.8 $\pm$ 9.9</u>
Random, 90%	93.0 $\pm$ 6.6	<u>83.2 $\pm$ 6.3</u>	91.0 $\pm$ 9.0	75.7 $\pm$ 13.3	<u>68.9 $\pm$ 6.3</u>	61.5 $\pm$ 4.0
Random (channel $\times$ time)*	92.6 $\pm$ 8.0	85.2 $\pm$ 4.0	82.8 $\pm$ 6.5	74.7 $\pm$ 14.3	73.6 $\pm$ 15.4	54.8 $\pm$ 9.9
Whole channels	95.4 $\pm$ 4.8	89.3 $\pm$ 8.1	94.8 $\pm$ 6.4	<u>74.0 $\pm$ 14.2</u>	<u>65.2 $\pm$ 9.5</u>	<u>53.5 $\pm$ 4.6</u>
Whole time patches	89.5 $\pm$ 13.2	84.0 $\pm$ 10.9	81.1 $\pm$ 11.4	63.7 $\pm$ 14.7	55.7 $\pm$ 4.2	48.3 $\pm$ 3.4
Brain regions	89.3 $\pm$ 11.4	82.8 $\pm$ 14.0	79.2 $\pm$ 3.4	67.3 $\pm$ 10.8	62.4 $\pm$ 4.7	52.1 $\pm$ 5.5
Last time patches (causal)	91.0 $\pm$ 9.2	75.2 $\pm$ 13.2	79.4 $\pm$ 7.2	63.2 $\pm$ 14.2	52.2 $\pm$ 3.8	50.0 $\pm$ 0.0

AD: Alzheimer’s disease; FTD: frontotemporal dementia; FEP: first-episode psychosis; all versus healthy controls. LP: linear probe. *: canonical setting (random masking, 75%), the configuration used for BrainLM-EEG in the main tables. Best and second best are ranked within each block. Values are subject level $\times 100$. **Bold**: best per column; underline: second best.

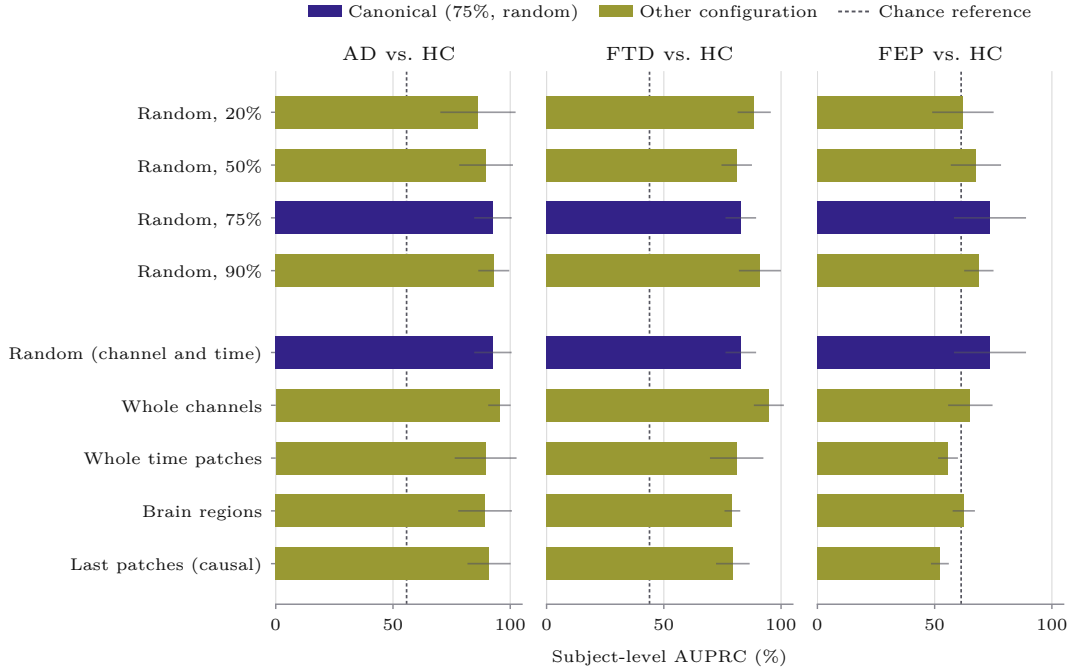


Figure 5.5: Subject level AUPRC for masking ratio and masking strategy ablations across tasks. The upper block varies the masking ratio under random masking; the lower block varies the masking structure at the canonical 75% ratio.

For the masking strategy comparison, the masking ratio was fixed at 75%. Channel-wise masking produced the highest AUPRC on both neurodegenerative tasks. On AD, it achieved 95.4% AUPRC and 89.3% BAcc, compared with 92.6% and 85.2% for random masking. On FTD, channel-wise masking achieved 94.8% AUPRC compared with 82.8% for random masking, while BAcc was similar (74.0% versus 74.7%). On FEP, random masking performed better than channel-wise masking on both reported metrics, including an AUPRC of 73.6% versus 65.2%. The remaining temporal, region-wise, and forecasting style masking strategies were below random masking in most reported columns. Figure 5.5 summarises both ablation blocks.

The masking ablation therefore shows that the pretraining configuration matters, but that its effect is task dependent. Channel-wise masking was favourable on several AD and FTD comparisons, whereas random masking remained stronger on FEP. No single masking configuration dominated all tasks and metrics. The result is consequently not a universal best masking rule, but evidence that the structure of the pretraining corruption changes the usefulness of the resulting representation.

5.7 Layer-Wise Representation Analysis

Having established that downstream performance depends on both task and pretraining configuration, the layer-wise analysis asks where task relevant information is expressed within BrainLM-EEG (Section 4.7.2). Table 5.7 reports results based on mean-pooled patch token representations rather than the CLS representation used in the main BrainLM-EEG analyses.

Table 5.7: Layer-wise linear probe analysis of BrainLM-EEG on frozen representations, pooling the mean over patch tokens (subject level).

	AD		FTD		FEP		MDD	
	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$
Layer 0	64.9 $\pm$ 14.4	50.0 $\pm$ 0.0	68.0 $\pm$ 9.7	50.0 $\pm$ 0.0	62.2 $\pm$ 14.2	50.0 $\pm$ 0.0	65.8 $\pm$ 12.3	50.0 $\pm$ 0.0
Layer 1	85.2 $\pm$ 10.7	<u>84.8 $\pm$ 11.0</u>	<u>90.0 $\pm$ 7.4</u>	78.5 $\pm$ 12.3	65.1 $\pm$ 7.3	<u>51.8 $\pm$ 6.8</u>	<u>58.0 $\pm$ 8.2</u>	50.0 $\pm$ 0.0
Layer 2	<u>91.9 $\pm$ 11.0</u>	83.3 $\pm$ 15.3	92.8 $\pm$ 6.8	<u>70.5 $\pm$ 11.6</u>	68.4 $\pm$ 13.4	58.8 $\pm$ 11.5	51.0 $\pm$ 7.0	50.0 $\pm$ 0.0
Layer 3	95.5 $\pm$ 4.2	87.0 $\pm$ 7.3	86.5 $\pm$ 9.0	60.2 $\pm$ 9.5	71.0 $\pm$ 10.6	51.1 $\pm$ 5.3	52.5 $\pm$ 7.4	48.6 $\pm$ 1.7
Final	91.2 $\pm$ 11.7	78.3 $\pm$ 10.6	81.6 $\pm$ 14.2	65.5 $\pm$ 24.2	<u>68.4 $\pm$ 9.0</u>	50.5 $\pm$ 5.3	52.9 $\pm$ 6.5	<u>49.3 $\pm$ 1.4</u>

AD: Alzheimer’s disease; FTD: frontotemporal dementia; FEP: first-episode psychosis; MDD: major depressive disorder; all versus healthy controls. Layer 0: patch embedding; Layers 1 to 3: transformer blocks; Final: encoder output (last block with final norm). Values are subject level $\times 100$. **Bold**: best per column; underline: second best.

For AD, layer 3 produced the strongest reported values, with 95.5% AUPRC and 87.0% BAcc. The final transformer layer achieved 91.2% AUPRC and 78.3% BAcc, below

layers 2 and 3. A similar pattern was observed for FTD, where layer 2 achieved 92.8% AUPRC compared with 81.6% at the final layer. For FEP, layers 2 and 3 also provided the strongest reported decoding performance. Within this model and evaluation setup, task relevant decoding performance did not consistently peak at the final transformer layer.

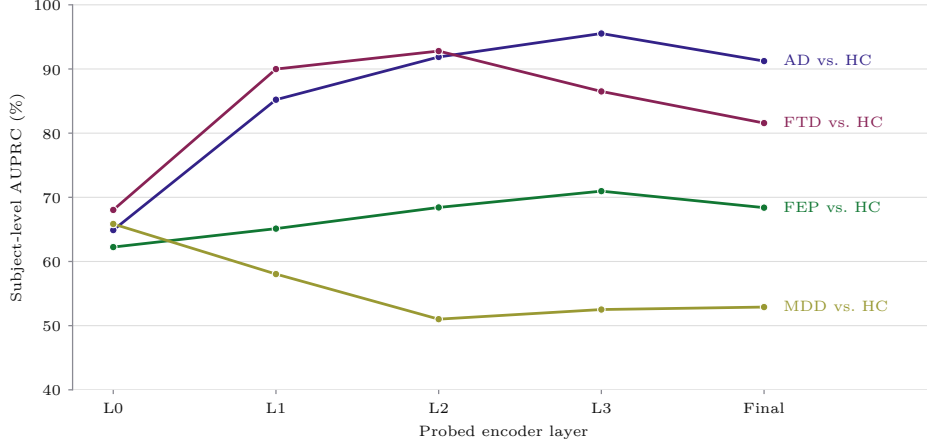


Figure 5.6: Mean-pooled patch token AUPRC across model layers for each task. AD and FTD peak at an intermediate layer, FEP increases gradually, and MDD declines after the patch embedding.

The MDD results differed across metrics. The pre-transformer patch embedding representation, reported as layer 0, achieved 65.8% AUPRC. AUPRC then decreased to 58.0%, 51.0%, 52.5%, and 52.9% across the transformer layers, while layer-0 BAcc was 50.0%. This is the opposite of the pattern for AD and FTD, where the values rise sharply from layer 0 to a peak at layer 3 or layer 2 before declining slightly toward the final representation. The layer-wise results again show a difference between ranking performance and thresholded classification for MDD. No transformer layer improved on the layer-0 MDD AUPRC, but this does not imply that the representation contains no decodable MDD information, because the reported AUPRC values remain above the 26.5% prevalence reference. Figure 5.6 shows these depth profiles for all four tasks.

Overall, the layer-wise analysis shows that the representation is structured across depth rather than becoming uniformly more useful toward the encoder output. Intermediate layers provided the highest decoding values for several tasks, while the final layer was not consistently best. This observation is descriptive and does not by itself identify why information changes across encoder depth.

5.8 CLS Token Contribution to Reconstruction

The layer-wise results concern mean-pooled patch tokens, whereas the main BrainLM-EEG downstream analyses use the CLS token as their global representation. Because this token has no token specific reconstruction target during masked autoencoder pretraining, the perturbation experiment described in Section 4.7.3 therefore tested whether changing the CLS representation affected reconstruction by the decoder. The held-out pretraining validation split contained 4,584 windows evaluated in 71 batches. For each batch, the encoder output and mask were held fixed while the decoder was run with the original CLS token, a zero vector, or Gaussian noise.

Perturbing the CLS token increased masked patch reconstruction MSE in all 71 paired batches. Mean batch level MSE increased from 0.4492 ± 0.0109 with the original token to 0.4666 ± 0.0110 after zeroing the token and 0.4691 ± 0.0112 after replacement with Gaussian noise. These changes correspond to increases of 3.86% and 4.42%, respectively. The one-sided paired Wilcoxon signed-rank tests gave $W = 2556$ and $p = 1.2 \times 10^{-13}$ for both perturbations at $\alpha = 0.05$. The batch level comparison is shown in Figure 5.7.

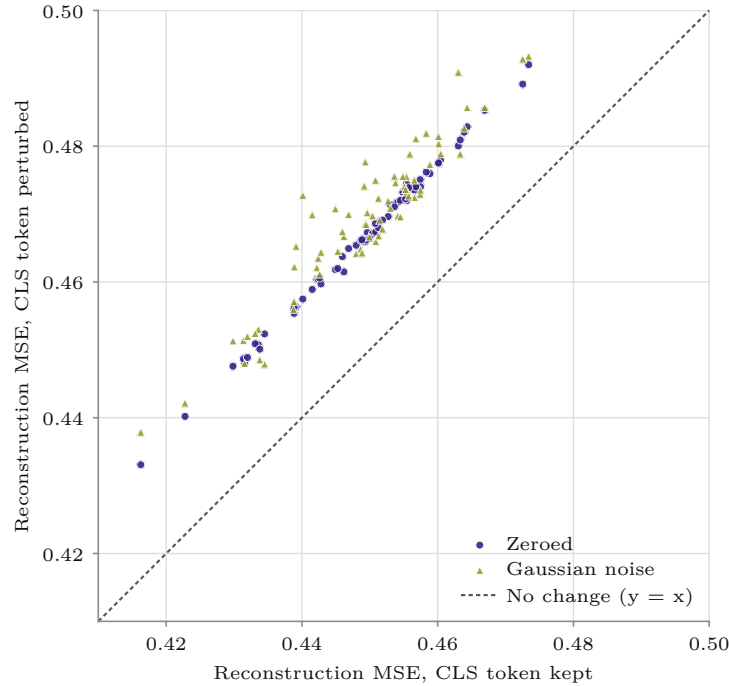


Figure 5.7: Masked patch reconstruction MSE with the original CLS token versus zeroed or Gaussian-noise replacement across held-out validation batches. The dashed line indicates equality; points above the line indicate increased MSE after perturbation.

The consistent increase in reconstruction error indicates that the decoder uses information associated with the CLS representation. However, the experiment does not quantify what

proportion of the reconstructed information is carried by CLS, nor does it show that this token contains clinically discriminative information. The statistical unit in this analysis was also the batch rather than the participant. The reported p -values therefore describe consistency across the evaluated batches and should not be interpreted as participant level inferential evidence.

5.9 Analysis of Reconstructed EEG

Table 5.8: BrainLM reconstruction quality assessment: classification from the input EEG versus from the BrainLM encode→decode output (subject level).

	AD		FTD		FEP	
	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$	AUPRC $\uparrow$	BAcc $\uparrow$
LDA (input EEG)	91.3 $\pm$ 10.9	94.3 $\pm$ 7.9	77.2 $\pm$ 9.0	65.8 $\pm$ 11.6	64.4 $\pm$ 10.5	58.2 $\pm$ 8.3
LDA (reconstructed EEG)	95.0 $\pm$ 7.1	89.0 $\pm$ 9.0	89.7 $\pm$ 7.5	81.7 $\pm$ 10.7	72.2 $\pm$ 6.0	65.1 $\pm$ 11.2
ATCNet (input EEG)	92.4 $\pm$ 9.5	83.2 $\pm$ 7.3	87.5 $\pm$ 10.4	71.0 $\pm$ 25.5	62.8 $\pm$ 5.6	56.5 $\pm$ 4.6
ATCNet (reconstructed EEG)	94.1 $\pm$ 7.8	80.7 $\pm$ 6.1	89.1 $\pm$ 8.3	65.0 $\pm$ 14.6	78.2 $\pm$ 5.0	51.1 $\pm$ 6.3

AD: Alzheimer’s disease; FTD: frontotemporal dementia; FEP: first-episode psychosis; all versus healthy controls. Reconstructed EEG is the BrainLM full-autoencoder output (mask ratio = 0) of the same input, with the classifier retrained on it. Values are subject level $\times 100$. **Bold**: better of the two rows for that classifier.

The final representation analysis moves from latent features back to the EEG signal and asks whether information useful for downstream classification is retained after reconstruction by BrainLM-EEG, following Section 4.7.4. LDA and ATCNet were each trained separately on original and reconstructed inputs for AD, FTD, and FEP. Table 5.8 reports the six resulting paired model task comparisons. MDD was not included in this analysis.

AUPRC was higher with reconstructed EEG in all six comparisons. The largest reported increases included LDA on FTD, from 77.2% to 89.7% AUPRC, and ATCNet on FEP, from 62.8% to 78.2% AUPRC. For LDA on FTD, BAcc also increased from 65.8% to 81.7%. However, BAcc did not improve consistently. It decreased for LDA on AD (94.3% to 89.0%), ATCNet on AD (83.2% to 80.7%), ATCNet on FTD (71.0% to 65.0%), and ATCNet on FEP (56.5% to 51.1%), while increasing for LDA on FTD and FEP (Figure 5.8).

The reconstruction experiment therefore complements the latent representation analyses. The reconstructed inputs produced higher ranking performance in all six reported AUPRC comparisons, while the effect on thresholded participant level classification was mixed. This shows that the encoder-decoder transformation retained information that the evaluated classifiers could use, even though the experiment does not establish that

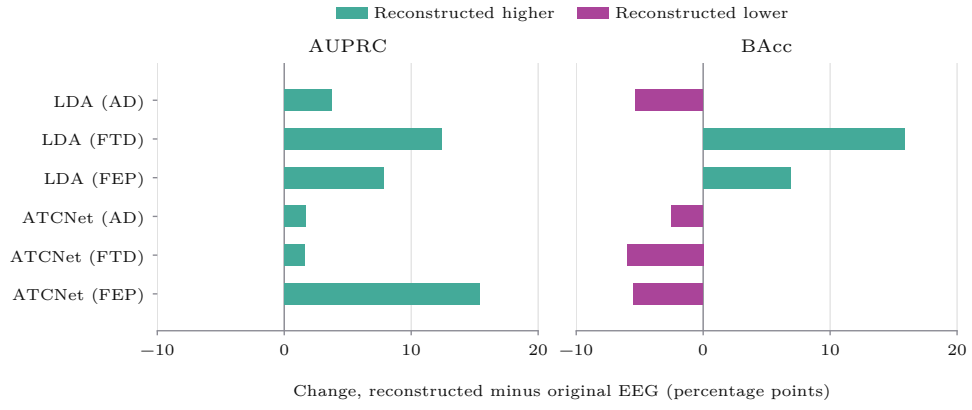


Figure 5.8: Change in classifier performance when using reconstructed versus original EEG inputs. Bars show reconstructed minus original performance for AUPRC and balanced accuracy across tasks.

reconstruction denoises the EEG or preserves all clinically relevant information. No inferential comparison or signal level analysis was performed, so the finding is limited to the reported classifiers and tasks.

6 Discussion

6.1 Synthesis of Findings

Taken together, the experiments reveal three broader findings. First, self-supervised pretraining produced resting-state EEG representations that supported participant level clinical classification, but their usefulness was task dependent. The proposed BrainLM-EEG variants were competitive across several tasks, with their clearest relative advantage observed on FTD under LOSO. Strong performance on AD across multiple model families further suggests that clinically relevant information in that dataset was accessible to a range of modelling approaches.

Second, within-dataset and cross-dataset generalisation were distinct challenges. Representations that transferred successfully to unseen participants within the same dataset did not retain the same level of performance on the independently acquired SCZ dataset. This indicates that participant level evaluation alone provides limited evidence of robustness to changes in cohort and acquisition conditions.

Third, the usefulness of the learned representations depended on design and extraction choices. Masking configuration and encoder depth affected downstream performance, while the CLS-token and reconstruction analyses showed that information relevant to reconstruction and classification was retained beyond a single final representation. The geometry-aware and microstate-informed variants produced improvements in individual settings, but neither provided a consistent advantage across tasks.

Overall, self-supervised pretraining produced reusable EEG representations that performed competitively across multiple clinical tasks and showed that their downstream utility was influenced by the clinical task, evaluation protocol, representation design, and degree of dataset shift. Consistent superiority over supervised or classical alternatives was not observed.

6.2 Task Dependence of the Learned Representations

A central finding is that the amount of participant level separation differed substantially across the four downstream tasks, while differences among model families were generally smaller and less consistent. Under within-dataset evaluation, balanced accuracy ranged from 67.8 to 87.3 on AD, from 48.5 to 81.3 on FTD, from 47.1 to 60.4 on FEP, and from 48.0 to 55.3 on MDD (Figure 5.1). These results show that the downstream task placed a stronger constraint on performance than model family alone. This pattern is important because the evaluated self-supervised representations did not provide the same level of downstream separability across clinical tasks. Their usefulness should therefore be interpreted in the context of the specific clinical task and evaluated cohort rather than assumed to transfer uniformly across applications.

The strongest evidence for useful pretrained representations comes from AD and FTD. On AD, high performance was obtained by several model families, including simple classical baselines. The contribution of this task is therefore not evidence of superiority of one architecture, but evidence that the learned representations preserve information that supports strong participant level classification. FTD is more informative for comparing representations. Under LOSO, the proposed BrainLM-EEG variants were among the strongest models, with BrainLM-EEG-RoPE reaching 92.3 AUPRC and 92.9 balanced accuracy and BrainLM-EEG-Microstate reaching 94.6 and 92.3. This shows that the proposed representation can support strong out-of-sample classification on a task for which the model ranking was not already saturated across all approaches.

FEP and MDD produced a different pattern. Their results were weaker and less consistent, but they should not be treated as null findings. For FEP, several models produced AUPRC values above the task reference, although the preferred model depended on the protocol. For MDD, whose evaluation pool contained 26 patients and 72 controls, LOSO AUPRC values ranged from 30.7 to 42.5 relative to the 26.5 prevalence reference, with LaBraM at 42.5 and EEGNeX at 41.9. Balanced accuracy ranged from 49.5 to 61.4 against the nominal chance level of 50. The result is therefore not that the task contained no usable information, but that the available ranking signal translated less consistently into thresholded participant level classification.

The layer-wise analysis strengthens this interpretation without requiring a claim about the biological origin of the task differences. For MDD, the highest AUPRC was observed at layer 0 at 65.8, followed by 58.0, 51.0, 52.5, and 52.9 through the transformer. No transformer depth improved MDD AUPRC beyond the patch embedding representation. For AD and FTD the opposite happened, AUPRC kept climbing as the transformer added layers, reaching 95.5 and 92.8, respectively. Because each task has a different chance level, these raw numbers are not directly comparable, but the direction of change is, for MDD, the score is highest before the transformer runs and gets worse afterwards, while for AD and FTD it is lowest before the transformer runs and gets better afterwards. The

relevant conclusion is therefore that the transformer increased downstream decodability much more clearly for AD and FTD than for MDD. The present experiments do not determine why the tasks differ, and the disease specific context for these differences has already been reviewed in Chapter 2.

6.3 Evaluation Protocol and Participant Level Generalisation

The comparison between five-fold cross-validation and LOSO provides an important methodological finding. The principal conclusion is not that one protocol reveals a uniquely correct model ranking, but that small clinical EEG cohorts can produce model rankings that depend strongly on how participants are resampled. The more reliable interpretation is therefore based on patterns that remain meaningful across evaluation settings rather than on small numerical differences within one split.

FTD provides the clearest example. Under five-fold cross-validation, BrainLM-EEG-RoPE reached 73.0 AUPRC and 48.5 balanced accuracy, below the nominal chance level of 50. Under LOSO, the same pretrained representation evaluated on the same 41 participants reached 92.3 and 92.9. BrainLM-EEG-Microstate similarly changed from 87.0 and 76.3 to 94.6 and 92.3 (Figure 5.2). With 41 participants, five-fold cross-validation results in both relatively small training sets and small test folds. The limited number of training participants can increase variability in the fitted model, while the small test folds make each individual participant influential on the fold level performance estimate. LOSO instead trains on nearly the full available cohort in each iteration, produces an out-of-sample prediction for every participant, and pools these predictions into a single participant level estimate.

The important finding is therefore twofold. First, the exact ordering of models on small cohorts should not be treated as stable when it changes with the resampling scheme. Second, the LOSO results still demonstrate that the proposed representations can generalise to held-out participants, particularly on AD and FTD. In this sense, the protocol comparison strengthens rather than weakens the thesis: it separates evidence that the representation contains useful participant level information from claims about small differences in model ranking.

6.4 Generalisation Across Datasets

The FEP-to-SCZ experiment extends the evaluation from unseen participants to an independently collected dataset. This distinction is central to the generalisation question. Good performance on held-out participants from the same dataset demonstrates participant level generalisation under a shared acquisition environment; external transfer additionally requires robustness to changes in cohort and acquisition conditions.

AUPRC decreased for every model when the FEP-trained models were evaluated on SCZ (Figure 5.3). Using the SCZ prevalence reference of 39.0 given in Table 5.1, the highest transfer AUPRC under the five-fold setting was 54.4 for ATCNet and the highest under the all-data setting was 57.6 for EEGNeX. The proposed variants remained competitive in the first transfer setting, with values of 53.1 and 52.7, while their AUPRC values in the all-data setting lay between 42.2 and 47.4. Across the transfer table, balanced accuracy ranged from 47.4 to 63.4 against the nominal chance level of 50. The same challenge was observed for externally pretrained models: REVE, LaBraM, and CBraMod reached 46.1, 48.5, and 43.0 AUPRC, respectively, under the five-fold transfer protocol.

This result provides a clear boundary to the within-dataset findings. The proposed models learned representations that were useful for participant level classification, but those representations did not provide a consistent robustness advantage when both the cohort and acquisition setting changed. Importantly, this was not unique to BrainLM-EEG. Models pretrained on substantially larger EEG corpora showed the same difficulty, indicating that the FEP-to-SCZ shift remained challenging across modelling approaches.

The transfer experiment should nevertheless be interpreted in the context of the source task. The best within-dataset FEP AUPRC was 81.0, approximately 19 points above the FEP prevalence reference of 61.3. Because the source task was less strongly separated than AD or FTD, the experiment cannot isolate how much of the external performance reduction arose from the source task difficulty and how much arose from dataset shift. The conclusion that can be drawn directly is that, for this shift pair, participant level generalisation did not translate into a consistent cross-dataset robustness advantage. An external evaluation anchored on a stronger source task would provide a useful next test of this distinction.

6.5 What the Representation Analyses Add

The analysis experiments provide evidence that goes beyond end-task model rankings. Together, the finetuning comparison, masking ablation, layer-wise analysis, CLS perturbation, and reconstruction experiment show how the pretrained representation can be

used and where its useful information is located.

The finetuning comparison indicates that, with the labelled cohort sizes available here, the frozen representation was already a useful basis for downstream classification. Finetuning did not consistently improve on linear probing under the evaluated configuration (Figure 5.4). This does not imply that finetuning is generally unsuitable. It does show that updating the full encoder was not necessary to obtain the strongest results observed here, and that the information learned during self-supervised pretraining could be accessed effectively using the lighter adaptation strategy. This is practically relevant for small clinical datasets, where unrestricted adaptation introduces many trainable parameters relative to the number of labelled participants.

The masking ablation provides the clearest evidence that the pretraining objective influences downstream representation quality. Masking whole channels rather than random spatiotemporal patches changed FTD AUPRC from 82.8 to 94.8 and changed AD AUPRC and balanced accuracy from 92.6 and 85.2 to 95.4 and 89.3 (Figure 5.5). The FTD difference is 12 points, while the AD changes are approximately 3 to 4 points. The advantage was not universal, because FEP favoured random masking, and each pretraining configuration was evaluated from a single run. The conclusion is therefore not that one masking strategy is optimal for every EEG task, but that masking structure is a substantive design choice, not a neutral implementation detail. Similarly, the results with masking ratios of 75% and the other evaluated settings show that the masking configuration inherited from vision models should be validated for EEG and not assumed to transfer unchanged.

The layer-wise experiment adds a second design insight. Downstream information did not simply accumulate until the final encoder output. The strongest AD result occurred at layer 3 and the strongest FTD result at layer 2, while the final representation was lower by 4 and 11 AUPRC points, respectively (Figure 5.6). This means that selecting the representation depth is itself part of downstream adaptation. For clinical EEG tasks, the final transformer layer should not automatically be assumed to be the most informative representation.

The CLS perturbation provides direct evidence that the global token participates in reconstruction. Replacing it increased reconstruction error in all 71 held-out validation batches, by 3.86% when zeroed and 4.42% when replaced by Gaussian noise (Figure 5.7). The effect is modest relative to the roughly 71 visible patch tokens available to the decoder, so the CLS token should not be interpreted as the sole information bottleneck. Nevertheless, the consistent increase establishes that it carries information used by the decoder rather than acting as an unused downstream placeholder.

Finally, the reconstruction experiment shows that the learned encoder-decoder preserves information that remains useful to classifiers with very different inductive biases. AUPRC increased when LDA and ATCNet were trained on reconstructed rather than original

EEG in all six reported comparisons, while balanced accuracy improved in only part of them (Figure 5.8). This is a particularly useful result because the effect is not tied to the downstream BrainLM classification head. It indicates that the reconstructed signal retains information relevant to participant ranking, as reflected by the higher AUPRC observed in the reported comparisons. The experiment does not establish a denoising mechanism, but it does provide independent evidence that the pretrained model learns a transformation that preserves clinically useful structure.

The architectural variants should be interpreted against these stronger representation level findings. BrainLM-EEG-Microstate and BrainLM-EEG-RoPE both produced strong results in individual settings, but neither modification showed a consistent improvement across tasks and protocols. This does not undermine the base representation learning approach. Instead, it indicates that the clearest gains in the present experiments came from choices such as masking and representation depth rather than from a universally superior positional or auxiliary loss modification. One possible explanation for the limited effect of the geometry-aware encoding is the common 19-electrode montage: with only 19 fixed spatial positions, a learned spatial embedding may already capture much of the available positional structure. This remains a hypothesis that should be tested with higher-density or variable-channel data.

6.6 Implications for Self-Supervised EEG Modelling

Several practical conclusions follow from the combined experiments. First, evaluation should distinguish between generalisation to unseen participants and robustness to a new dataset. The former was demonstrated on the stronger downstream tasks, whereas the latter remained difficult in the FEP-to-SCZ experiment. Second, small clinical cohorts require care when model rankings are interpreted, because the FTD results show that rankings can change substantially with the resampling protocol. Third, the pretrained representation should not be treated as a fixed final layer object: masking strategy, representation depth, and adaptation method all affected how much useful information could be extracted.

These findings also clarify where self-supervised pretraining contributed most in the present study. Its value was strongest as a reusable representation learning stage. The frozen encoder supported competitive downstream performance, intermediate layers contained highly decodable information, and reconstructed EEG remained informative to independent classifiers. The results therefore support the use of self-supervised learning as a way of learning general EEG structure before labelled downstream adaptation, while showing that successful deployment still requires task specific choices about evaluation, representation extraction, and external validation.

6.7 Limitations, Threats to Validity and Reproducibility

The limitations affect the precision with which individual model differences can be interpreted, but they do not remove the broader patterns observed across the experiments. The first limitation is statistical. No significance testing was performed on the downstream model comparisons, and no confidence intervals were computed. Each pretraining configuration was also run with a single seed, so small differences between variants and masking configurations cannot be separated from run-to-run variation. The conclusions therefore focus on recurring patterns and larger directional changes rather than treating small numerical leads as evidence of general superiority.

The second limitation concerns the data. The downstream cohorts contain between 41 and 98 subjects, which is small relative to the number of parameters in the models. Only resting-state recordings were used, and all datasets were reduced to the common 19-channel montage. This standardisation supports comparison across datasets but sacrifices spatial resolution. Cross-dataset generalisation was assessed using a single external cohort, so the transfer conclusion is specific to one source-target pair.

The third limitation concerns the models and adaptation protocols. Finetuning used a single global learning rate with no layer-wise decay, no partial unfreezing, and no per-task search. Its lower performance relative to linear probing is therefore specific to that recipe. The classification head used for probing is a three-layer perceptron rather than a strictly linear classifier, so the relevant interpretation is downstream decodability, not linear separability. Comparisons with externally pretrained foundation models are also comparisons between complete pipelines, since those models differ in pretraining corpus, tokenisation, architecture, and input processing.

Several aspects of the experimental design strengthen the internal validity of the findings. All twelve models were evaluated on identical participant level partitions using the same aggregation and metric implementation. The partition between pretraining and downstream evaluation was made at the participant level, and the pretraining reserve mechanism described in Section 4.1 prevented evaluation participants from entering the self-supervised pretraining corpus. These controls reduce the risk that the observed downstream performance is explained by participant level leakage.

6.8 Answers to the Research Questions

The four research questions stated in Section 1.1 can now be answered directly by considering the experiments together.

For the first research question, self-supervised pretraining produced useful downstream EEG representations. The proposed models were competitive with supervised, classical, and externally pretrained alternatives across several within-dataset evaluations and were among the strongest approaches on FTD under LOSO. They did not consistently outperform every baseline, so the evidence supports the usefulness and competitiveness of the learned representation rather than universal model superiority. Linear probing was the more effective adaptation strategy under the evaluated cohort sizes and hyperparameters.

For the second research question, the experiments demonstrate participant level generalisation but identify a clear distinction between within-dataset and cross-dataset robustness. LOSO produced out-of-sample predictions for every participant and showed strong performance on the best-performing tasks. In the FEP-to-SCZ evaluation, however, AUPRC decreased for every model relative to its within-dataset value, and no model family showed a consistent robustness advantage. Cross-dataset robustness therefore remains a more demanding problem than participant level generalisation within a shared dataset.

For the third research question, the experiments show that representation quality depends on pretraining and representation extraction choices. Masking ratio, masking strategy, and encoder depth all changed downstream performance. Intermediate layers produced the strongest decoding values for several tasks, and the canonical 75% random masking setup was not uniformly best. The RoPE and microstate modifications did not produce a stable advantage across evaluation protocols, indicating that their benefit remains task dependent under the present setup.

For the fourth research question, the representation analyses provide direct evidence that BrainLM-EEG learns structured and reusable information. The CLS token contributes to reconstruction, task relevant information is distributed non-uniformly across encoder depth, and reconstructed EEG produced higher AUPRC in all six reported classifier comparisons. These findings show that the learned representation affects both latent space decodability and the information available after reconstruction.

Overall, the thesis demonstrates that self-supervised pretraining is a viable representation learning strategy for resting-state EEG and that its value is most clearly expressed through reusable participant level representations rather than through universal superiority of one architecture. The experiments identify concrete design choices that influence those representations, show that the learned information can support strong downstream classification in selected clinical tasks, and distinguish successful within-dataset generalisation from the more difficult problem of external dataset transfer. These conclusions provide a clear basis for improving future EEG foundation models: stronger external validation, task-aware masking, representation depth selection, and adaptation strategies matched to the size of the labelled cohort.

7 Conclusion

This thesis investigated whether self-supervised pretraining can produce useful and generalisable representations of resting-state EEG for clinical classification. BrainLM, originally developed for fMRI, was adapted to multichannel EEG and evaluated across four clinical classification tasks. Two controlled extensions were also investigated, incorporating electrode geometry and EEG microstates, together with the effects of masking configuration, encoder depth, and downstream adaptation.

Overall, the results provide a conditional assessment of self-supervised EEG representation learning. The proposed BrainLM-EEG models produced reusable representations that were competitive with classical machine learning, supervised deep learning, and existing pretrained EEG models. Performance was strong in absolute terms on AD and FTD, with the clearest relative advantage observed for FTD under LOSO evaluation. However, no model or architectural modification was consistently superior across tasks and evaluation settings. The benefit of self-supervised pretraining therefore depended on the clinical task, the encoder representation used for downstream classification, and the evaluation protocol.

The results further demonstrate an important distinction between participant level and cross-dataset generalisation. Performance on held-out participants within the same dataset did not necessarily translate to an independently acquired dataset. In particular, the FEP-to-SCZ transfer experiment showed that differences in cohorts and acquisition conditions remained challenging for both pretrained and supervised approaches. Strong within-dataset performance should therefore not be interpreted as evidence of robustness to broader dataset shifts.

The representation analyses showed that downstream performance was also influenced by choices made during pretraining and representation extraction. Masking configuration and encoder depth affected classification performance, and intermediate encoder layers were more informative than the final layer for some tasks. CLS-token perturbation indicated that the token contributed to masked reconstruction, while reconstructed EEG retained information that could be used for downstream classification. The geometry-aware and microstate-informed variants produced improvements in individual settings, but neither provided a consistent advantage across tasks.

Taken together, these findings support masked self-supervised learning as a viable

approach for learning reusable representations from resting-state EEG, while showing that representation quality and downstream utility depend on the clinical task, model design, and evaluation setting. The main challenge is therefore not only to learn informative EEG representations, but to understand when they are useful and whether that utility generalises across participants and independently collected datasets. Further progress will require stronger external validation, careful control of dataset and participant effects, and model designs that account for the substantial heterogeneity of clinical EEG.

7.1 Future Work

First, cross-dataset evaluation should be extended to additional independent EEG datasets. The present external evaluation was limited to FEP-to-SCZ transfer. Testing additional source-target pairs, particularly for AD and FTD where stronger within-dataset performance was observed, would provide stronger evidence of whether the learned representations remain useful across different cohorts and acquisition settings.

Second, differences in the usefulness of the learned representations across clinical conditions should be investigated more closely. AD and FTD showed stronger and more consistent downstream performance than FEP and MDD, while the layer-wise analysis also revealed different patterns across tasks. Future work should investigate which signal characteristics, learned representations, and task formulations contribute to these differences. This could help determine whether alternative representations or modelling strategies are better suited to clinical tasks for which the present approach produced weaker or less consistent results.

Third, future work could examine individual deviations from a healthy reference distribution in the learned EEG representation space. Rather than formulating every clinical task as a direct patient-versus-control classification problem, representations from healthy participants could be used to characterise typical variation, against which individuals from clinical groups could be compared. This would shift the analysis towards how and to what extent an individual’s EEG representation differs from the reference population, which may be particularly useful for heterogeneous neurological and psychiatric conditions. Such an approach could also be used to investigate whether these individual deviation patterns remain stable across independently collected datasets.

Several evaluation choices could also be examined more systematically. Participant level probabilities were obtained by arithmetic averaging across EEG windows; alternative aggregation rules, including geometric averaging, could be evaluated to determine how sensitive performance is to this choice. The substantial variation observed under five-fold cross-validation and the protocol-dependent changes in model ranking indicate uncertainty in the relative performance of individual models. Future work could therefore assess

whether observed differences between models are robust to participant sampling, for example by resampling the paired out-of-sample participant predictions and estimating the uncertainty of the performance difference. Finally, the five FEP-trained cross-validation models were evaluated separately on the external SCZ dataset and their metrics were averaged. Combining their participant level predictions through probability averaging or voting would provide an additional ensemble evaluation and could determine whether aggregation across independently fitted models improves cross-dataset robustness.

Together, these directions extend the present work towards stronger cross-dataset validation, more robust model comparison, a better understanding of task specific representation differences, and more individual-level analysis of clinical EEG.

7.2 Personal Takeaways

This thesis gave me the opportunity to work on a research problem at the intersection of Artificial Intelligence and neuroscience. One of the most valuable lessons was learning how to connect these two areas in a meaningful way. With the guidance of my supervisors, I learned to look beyond model performance and consider how factors such as preprocessing, dataset characteristics, evaluation design, and the clinical task can shape the results. This helped me develop a broader understanding of how AI methods can be applied to neuroscience problems and why their results need to be interpreted in the context of both the data and the application.

The research process also taught me how to develop a broad idea into focused research questions and a structured set of experiments. Reading and comparing previous studies helped me become more critical about methodological choices and more careful about what can reasonably be concluded from experimental results. I also learned that research does not always lead to clear or expected outcomes. Some experiments produced strong results, while others were less consistent, and learning how to understand these differences and draw appropriate conclusions without overclaiming was an important part of the process.

Another important learning experience was managing a long-term research project involving multiple datasets, model configurations, experiments, analyses, and writing tasks. I learned how to plan the work in stages, prioritise tasks, manage dependencies, and adapt the research direction as new results emerged. Regular discussions and feedback from my supervisors also helped me improve the way I justify technical decisions, communicate results, and structure scientific arguments. Overall, the thesis strengthened my ability to approach complex interdisciplinary research problems in a more systematic and independent way.

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List of Abbreviations

AD	Alzheimer’s disease
ASR	Artifact Subspace Reconstruction
AUPRC	Area under the precision-recall curve
AUROC	Area under the receiver operating characteristic curve
BAcc	Balanced accuracy
BCI	Brain-computer interface
CLS	Classification token
CNN	Convolutional neural network
EC	Eyes closed
EEG	Electroencephalography
EO	Eyes open
FEP	First-episode psychosis
fMRI	Functional magnetic resonance imaging
FN	False negative
FP	False positive
FT	Full finetuning
FTD	Frontotemporal dementia
GELU	Gaussian error linear unit
HC	Healthy control
ICA	Independent component analysis
LDA	Linear discriminant analysis
LOSO	Leave-one-subject-out
LP	Linear probe
LSTM	Long short-term memory
MAE	Masked autoencoder
MDD	Major depressive disorder
MSE	Mean squared error
RoPE	Rotary positional encoding
SSL	Self-supervised learning
SVM	Support vector machine
TN	True negative
TP	True positive

Datasets

ADFTD	Alzheimer's disease and frontotemporal dementia dataset
DVS	Dortmund Vital Study
FEP	First-episode psychosis dataset
LEMON	Leipzig Study for Mind-Body-Emotion Interactions
MDD	Major depressive disorder dataset
SCZ	Schizophrenia dataset
SRM	SRM resting-state EEG dataset

Appendix

This appendix provides the implementation settings used for the downstream experiments reported in Chapter 5.

A.1 Downstream Adaptation

Table A.1 shows the optimisation settings used for linear probing and finetuning. Within each adaptation setting, the same hyperparameters were used for all four downstream tasks, all three BrainLM-EEG variants, and all evaluation protocols.

Table A.1: Optimisation settings for linear probing and finetuning.

Setting	Linear probing	Finetuning
Encoder parameters	Frozen	Updated
Learning rate	1×10^{-3}	5×10^{-5}
Weight decay	0	0.05
Learning rate schedule	Cosine decay	Linear warmup over 10 epochs, followed by cosine decay
Maximum epochs	50	50
Early stopping patience	10	10
Batch size	128	128
Gradient clipping (maximum norm)	1.0	1.0

All runs used the AdamW optimiser (Loshchilov and Hutter, 2019), a random seed of 42, and cross-entropy loss. Early stopping and checkpoint selection were based on validation loss.

The classification head described in Section 4.6.2 was a three-layer perceptron:

$$\text{CLS} \rightarrow \text{Linear}(512, 1024) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.1) \rightarrow \\ \text{Linear}(1024, 1024) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(1024, 2)$$

The same classification head was used in both settings. The encoder remained frozen during linear probing and was updated during finetuning.

All downstream classification results were obtained from the CLS token. The layer-wise analysis in Section 4.7.2 was the only exception. For this analysis, the patch token representations from each encoder stage were averaged, and the CLS token was excluded. Therefore, the results in Section 5.7 are not directly comparable with the CLS-based results reported elsewhere.

A.2 Electrode Montage and Coordinates

Channel names were standardised to the current electrode nomenclature. The legacy labels T3, T4, T5, and T6 were mapped to T7, T8, P7, and P8, respectively.

The channels were ordered from anterior to posterior and from left to right within each row. This order was kept fixed so that each spatiotemporal token corresponded to the correct electrode.

The azimuth and elevation angles used by BrainLM-EEG-RoPE (Section 4.3.2) were derived from the standard 10-20 electrode coordinates provided by `MNE-Python`. The coordinates use a head-centred Cartesian system. The x axis points towards the right ear, the y axis points towards the nasion, and the z axis points upwards.

Each electrode position was projected onto the unit sphere. Azimuth and elevation were then calculated as

$$\theta_c = \text{atan2}(x_c, y_c), \quad \varphi_c = \arcsin(z_c) \quad (\text{A.1})$$

Table A.2 lists the electrode order and the corresponding coordinates.

Table A.2: The 19 common electrodes in the order used by the model, with their spherical coordinates.

Index	Electrode	Azimuth θ_c	Elevation φ_c	Index	Electrode	Azimuth θ_c	Elevation φ_c
0	Fp1	-0.337	-0.078	10	C4	+1.732	+0.752
1	Fp2	+0.338	-0.079	11	T8	+1.746	-0.109
2	F7	-1.027	-0.138	12	P7	-2.363	-0.024
3	F3	-0.758	+0.523	13	P3	-2.549	+0.532
4	Fz	+0.005	+0.849	14	Pz	+3.138	+0.795
5	F4	+0.762	+0.498	15	P4	+2.525	+0.531
6	F8	+1.024	-0.139	16	P8	+2.356	-0.025
7	T7	-1.759	-0.109	17	O1	-2.886	+0.076
8	C3	-1.747	+0.770	18	O2	+2.882	+0.076
9	Cz	+3.098	+1.480				

Azimuth θ_c and elevation φ_c are given in radians. These values were used directly as the scalar positions in Equation 4.9. Rotation frequencies used the base $b = 10^4$, without additional rescaling.

A.3 Code Availability

The implementation, configuration files, and scripts used in this thesis are publicly available at <https://github.com/gokul-pv/ssl-eeg>.

Declaration on Oath

I hereby certify that I have written my master thesis independently and have not yet submitted it for examination purposes elsewhere. All sources and aids used are listed, literal and meaningful quotations have been marked as such.

A handwritten signature in black ink, appearing to read 'Gokul', is positioned above a horizontal line.

1st September 2026, Gokul Perumbayil Vijayakrishnan

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Generative AI was used during the preparation of this thesis solely for language editing, including corrections to grammar, spelling, clarity, phrasing, and sentence structure. All technical and scientific content, analyses, interpretations, results, and conclusions were written and developed by the author.

OpenAI (2026). *ChatGPT* (Version GPT-5.6) [Large language model]. <https://chatgpt.com/>



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